PROBABILITY

SYNOPSIS

➢ **Random Experiment or Trial (R.E) :**
A repeatable experiment the results of which are known in advance but exact result will be known only after the experiment is completed is known as random experiment or an experiment, result of which can not be predicted with certainity is called random experiment.
a) Tossing a coin
b) Rolling a die
Drawing a specified number of cards from a well shuffled pack of 52 playing cards etc.,

➢ **Sample Space (S) :** The list of all possible outcomes or results of a random experiment is called as sample space for that experiment and it is denoted by S.

Ex: i) If a coin is tossed once, the sample space S is given by S = {H,T}
ii) If 2 coins are tossed once, the sample space S={HH,HT,TH,TT}
iii) If a die is rolled once the sample space S={1,2,3,4,5,6}

➢ **Elementary event or Indecomposable event:** The possible outcome or the result of a random experiment that can not be split further is called an elementary or simple event.

➢ **Event:** An outcome or the result of a random experiment is called an event.

Ex: While tossing a coin we may get head or tail upwards. Now getting head upwards is an event or getting tail upwards is also an event.

Ex: The event of getting head upwards is an elementary event.

➢ **Composite event :** The possible outcome or the result of a random experiment that can be further split into more than one elementary event is called a composite event.
The event can be either an elementary event or a composite event.

Ex: Getting a face with odd number of points upwards in rolling of a die is a composite event, since it can be divided further into three elementary events namely:

➢ **Mutually exclusive event (M.E.E.):**
The events of a R.E. are said to be M.E. events, if the occurence of any one of them prevents (or) eliminates the occurence of all the other events of that R.E.
Two events A and B are said to be M.E. or disjoint when both cannot happen simultaneously in a single trial or experiment. i.e., $A \cap B = \phi$.

Ex: In tossing a coin head turning upwards eliminates tail turning upwards in that particular R.E. So getting head & tail are M.E. events.
The events $A_1, A_2, A_3, ..., A_n$ of a R.E. are said to be M.E. or disjoint events if, $A_i \cap A_j = \phi$, for $i \neq j$ & $1 \leq i, j \leq n$.
Note : Whenever it is possible for two or more number of events to happen simultaneously, then those events are said to be compatible.

➢ **Equally likely events (E.L.E.):** The events of a R.E. are said to be E.L. When one does not occur more number of times than any one of the other events i.e., if all the events occur same number of times.
The events of a R.E. are said to be equally likely when there is no reason to expect a particular event in preference to any other event.

Ex: If an unbiased coin is tossed a large number of times, each face i.e., head and tail can be expected to appear same number of times.

➢ **Exhaustive events (E.E.):** The events of a random experiment are said to be exhaustive, if the occurence of any one of them is a certainity.
A set of events is said to be exhaustive, if these include all possible outcomes of the R.E.

Ex: In tossing a coin it is certain that either the head or the tail will turn upwards. So head & tail put together are called as exhaustive events.

➢ **Complementary Events :** Suppose A is any event of a random experiment associated with sample space S.
The complementary event of an event A denoted by $\overline{A}$ or A^C is the event given by $A^C = (S - A)$

Ex: If two coins are tossed, S={HH,HT,TH,TT}

Let A: occurence of atleast one head

Here A^C is the event of non - occurrence of atleast one head i.e, A^C is the event of getting both tails.

Here A={HH,HT,TH}, A^C={TT}

Note : Let S be a sample space. The event ϕ is called impossible event and the event S is called certain event in S.

➢ **Favourable Cases:** The possible outcomes or results of a R.E. which leads to the occurence of the event A are called as favourable cases for the event A to happen.

Ex: In rolling a symmetrical die 1, 3 & 5 are happening for the event A, the event of getting odd number of points upwards. So they are called as favourable cases for the event A.

➢ **Classical Definition of Probability (or) Mathematical Definition of Probability (or) A priori Probability:** The probability of the event A, denoted by P(A) is defined as $P(A)=\frac{m}{n}=\frac{n(A)}{n(S)}$.

Where n = The total number of possible outcomes in a R.E. which are M.E., E.L. & collectively exhaustive, m = The number of possible outcomes favourable to A in that R.E., n(S) = The total number of sample points in a sample space S, n(A) = The number of sample points favourable to A in that sample space S.

p(A) = $\frac{m}{n}$ can also be expressed as "The odds in favour of the event A are m to n - m or $\frac{m}{n-m}$" or "The odds against the event A are n-m to m or $\frac{n-m}{m}$". But in both the cases

$$P(A)=\frac{\text{favourable cases}}{\text{total no. of cases}}=\frac{m}{m+n-m}=\frac{m}{n}$$

Now consider $P(A)=\frac{m}{n}$

If m = 0, P(A) = 0, then A is said to be an impossible event.

If m=n, $P(A)=\frac{n}{n}=1$, then A is said to be a sure or certain event.

$\therefore$ The range of P(A) is [0, 1] i.e., $0\le P(A)\le 1$

From these we can conclude that the probability of any event is a non-negative rational number which lies between 0 & 1, both inclusive.

W.E.1:- ***A coin is tossed 2 times the probability of getting one head and one tail is***

Sol: Sample space, S = {HH, HT, TH, TT}; n(S) = 4

Let A be the event of getting 1 head and 1 tail

A = {HT, TH} n(A) = 2 $\therefore p(A)=\frac{n(A)}{n(S)}=\frac{2}{4}=\frac{1}{2}$

W.E.2:- ***Two dice are thrown simultaneously. The Probability of getting even numbers on both the dice is***

Sol: **n(S) =** $6^2=36$.

Let A be the event of getting even numbers on both the dice A = { (2,2), (2,4), (2,6), (4,2), (4,4), (4,6), (6,2), (6,4), (6,6)} ; n(A) = 9

$\therefore p(A)=\frac{n(A)}{n(S)}=\frac{9}{36}=\frac{1}{4}$

➢ A non-leap year = 52 weeks + 1 day

A leap - year = 52 weeks + 2 consecutive days of a week.

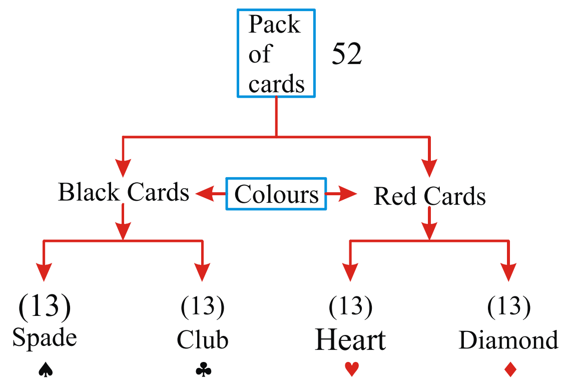
WE.3:- ***The probability that a non-leap year will have 53 Fridays is***

Sol: n(A)=1;n(S)=7 ; p(A)=n(A)/n(S)=1/7

W.E.4:- ***The probability that a leap year will have only 52 Sundays is***

Sol: n(A)=5;n(S)=7 ; p(A)=n(A)/n(S)=5/7

➢ **Description of Pack of Cards:**



➢ Here clubs, spades, diamonds, hearts are four suits.

- Each suit contains 13 cards namely 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A.
- In every suit J, Q, K are face cards (or court cards)
- In every suit J, Q, K, A are called honour cards.
- Jack card (J) is also called knave.
- The cards bearing the digits 2,3,4,5,6,7,8,9,10 are known as number cards. Ace is treated as 1.
- Each suit contains 9 numbered cards.
- In a pack of cards the number of cards with same face value is 4.
- **Odds in favour of A :** It is defined as $P(A):P(\bar{A})$

  Odds against A : It is defined as $P(\bar{A}):P(A)$

  If $P(A):P(\bar{A})=x:y$ then

  $P(A)=\dfrac{x}{x+y}$ and $P(\bar{A})=\dfrac{y}{x+y}$
- **Limitations:**
  If n is very very large, we can not determine the probability of the event A with the help of this definition.
  The probability can not be found when the outcomes are not equally likely.
  When the outcomes are not M.E., our logic may go wrong.
  One of the serious drawbacks of this definition is that in defining probability we use the term equally likely i.e., equally probable i.e., probability itself.
- All the identical trials are to be conducted independently. Again to define the concept of independence, we require the concept of probability.
- **Axiomatic approach to Probability :**
  Let S be a finite sample space associated with a random experiment. Then a real valued function P from the power set of S i.e., P(S) into the real line R is called a probability function on S, if P satisfies the following three axioms.
- **Axiom of Positivity:** For every subset A of S,

  $P(A)\ge 0$ i.e., $P(A)\ge 0, \forall A\in S$
- **Axiom of Certainty :** P(S) = 1
- **Axiom of Union or Axiom of Additivity:**
- If A_1, A_2 are two disjoint subsets of S then $P(A_1\cup A_2)=P(A_1)+P(A_2)$.

  Now the image P(A) of A is called the probability of the event A.

Note: In general, if $A_1, A_2, A_3, ..., A_n$ are 'n' disjoint subsets of S then $P(A_1\cup A_2\cup A_3\cup ......\cup A_n)$

$=P(A_1)+P(A_2)+P(A_3)+....+P(A_n)$

i.e., $P\left(\bigcup_{i=1}^{n} A_i\right)=\sum_{i=1}^{n} P(A_i)$

- If S consists of 'n' equally likely elementary events and A_i is one such elementary event of S, then

  $P(A_i)=\dfrac{1}{n}; i=1,2,3,......n$.
- **Notation :** Let A and B be two events in a sample space S.

Set theoric description	Events Related to Random experiment
1) $A\cup B$	Either the event A or B occurs (or) atleast one of A,B occurs
2) $A\cap B$	both the events A and B occur
3) $(A\cup B)^C = A^C\cap B^C$	neither A nor B occur
4) $(A\cap B)^C = (A^C\cup B^C)$	either A does not happen or B does not happen
5) $A\cap B^C$	A occurs but B dose not occur
6) $(A\cap B^C)\cup(A^C\cap B)$ $=(A-B)\cup(B-A)$ $=(A\cup B)-(A\cap B)$	Exactly one of A and B occur or Symmetric difference between the events A and B

- **Addition Theorem :** If A and B are any two compatible events in a sample space S, then the probability of occurence of either the event A or B or both the events A & B i.e., the event of $A\cup B$ is given by

  $P(A\cup B)=P(A)+P(B)-P(A\cap B)$ ....(1)

  If A and B are M.E. events, $P(A\cap B)=0$, then

  $P(A\cup B)=P(A)+P(B)$ ....(2)

  From equations (1) & (2) we can conclude that

  $P(A\cup B)\le P(A)+P(B)$

  Further $A, B\subseteq A\cup B$

  $P(A)$ or $P(B)\le P(A\cup B)\le P(A)+P(B)$

- If A and B are two events such that $A \subseteq B$, then $P(A) \leq P(B)$
- If A, B are two events of S then $P(A \cap B) \leq P(A) \leq P(A \cup B) \leq P(A) + P(B)$
- P(exactly one of A, B occurs)

$$= P\left(A \cap \overline{B}\right) + P\left(\overline{A} \cap B\right)$$

$$= P(A) + P(B) - 2P(A \cap B)$$

$$= P(A \cup B) - P(A \cap B)$$

$$= P\left(\overline{A} \cup \overline{B}\right) - P\left(\overline{A} \cap \overline{B}\right)$$

- If A, B are two events then

i) $P\left(\overline{A} \cup \overline{B}\right) = 1 - P(A \cap B)$

ii) $P\left(\overline{A} \cap \overline{B}\right) = 1 - P(A \cup B)$

- $\left(A \cap \overline{B}\right) \cup B = A \cup B$ & $\left(\overline{A} \cap B\right) \cup A = (A \cup B)$
- For three events A, B and C

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B)$$
$$-P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

- If A, B and C are M.E. events, the probabilities of compound events will be zero, then $P(A \cup B \cup C) = P(A) + P(B) + P(C)$
- If A, B and C are three events then P(exactly one of A, B, C occurs) =

$$P(A) + P(B) + P(C) - 2P(A \cap B)$$
$$-2P(A \cap C) - 2P(B \cap C) + 3P(A \cap B \cap C)$$

- P(exactly two of A, B, C occur) $= P(A \cap B)$

$$+P(A \cap C) + P(B \cap C) - 3P(A \cap B \cap C)$$

- P(at least two of A, B, C occurs) $= P(A \cap B)$

$$+P(A \cap C) + P(B \cap C) - 2P(A \cap B \cap C)$$

- If A,B and C are mutually exclusive and exhaustive events of the sample space S, then $A \cup B \cup C = S$ and P(A)+P(B)+P(C)=P(S)=1

W.E.5:- ***A die thrown. Let A be the event that the number obtained is greater than 3. Let B be the event that the number obtained is less than 5. Then $P(A \cup B)$ is (AIEEE 2008)***

Sol: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$= \frac{3}{6} + \frac{4}{6} - \frac{1}{6} = \frac{6}{6} = 1$$

W.E.6:- ***In a simulataneous throw of two dice, the porbability of A or B, if A=a sum of 11 points; B=an odd number of points on each die***

Sol : P (sum 11 or an odd no of points on each die)

$$= \frac{2}{36} + \frac{3}{6} \times \frac{3}{6} = \frac{11}{36}$$

- **Conditional Event:** If A,B are two events in a sample space then the event of happening of B after the happening of event A is called conditional event . It is denoted by B/A
- **Conditional Probability:** If A & B are two events in a sample space S such that $P(A) \neq 0$, then the probability of occurence of B, after the occurence of the event A is called the conditional probability of B given A, denoted by either $P(B/A)$ or $P\left(\frac{B}{A}\right)$ or $P(B;A)$ and is defined as

$$P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{n(A \cap B)}{n(A)}$$

Similarly if $P(B) \neq 0$, then

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B)}{n(B)}$$

In general $P(B/A) \neq P(A/B)$

W.E.7:- ***One ticket is selected at random from 50 tickets numbered 00,01,02,.....,49. Then the probability that the sum of the digits on the ticket is 8, given that the product of the digits is zero is (AIEEE 2009)***

Sol: E_1 = Event that the product of the digits is zero.

E_2 = Event that the sum of the digits is 8.

Favourable cases to

$E_1 = \{00, 01, 02, 03, ..., 09, 10, 20, 30, 40\}$

$E_2 = \{08, 17, 26, 35, 44\}$, $E_1 \cap E_2 = \{08\}$

$n(E_1) = 14$, $n(E_2) = 5$, $n(E_1 \cap E_2) = 1$

$\therefore$ Required probability

$$= P\left(\frac{E_2}{E_1}\right) = \frac{n(E_1 \cap E_2)}{n(E_1)} = \frac{1}{14}$$

W.E.8:- ***Three numbers are chosen at random without replacement from {1,2,3,....,8}. The probability that their minimum is 3, given that their maximum is 6, is [AIEEE 2012]***

Sol: E_1 = Event that the maximum number is 6

E_2 = Event that the minimum number is 3

Favourable cases to $E_1 = {}^5C_2$ (since 6 should be among the three and the rest two from 1 to 5)
Among favourable cases to E_1,

$(3,4,6)$ and $(3,5,6)$ are favourable to E_2.

$$\therefore P\left(\frac{E_2}{E_1}\right)=\frac{n(E_1\cap E_2)}{n(E_1)}=\frac{2}{{}^5C_2}=\frac{1}{5}$$

- **Multiplication Theorem (or) Product Theorem (or) Theorem of compound Probability:** If A and B are two events in a sample space S such that $P(A)\neq 0$ & $P(B)\neq 0$, then the probability of simultaneous occurence of the two events A and B is given by $P(A\cap B)=P(A).P(B/A)=P(B).P(A/B)$.
- For three events A,B and C $P(A\cap B\cap C)$ $=P(A).P(B/A).P(C/A\cap B)$
- For four events A, B, C and D $P(A\cap B\cap C\cap D)$ $=P(A).P(B/A).P(C/A\cap B).P(D/A\cap B\cap C)$

W.E.9:- ***Suppose there are 12 boys and 4 girls in a class. If we choose three children one after another in succession at random, find the probability that all the three are boys.***

Sol: Let A be the event of choosing a boy in 1st trial
Let B be the event of choosing a boy in 2nd trial
Let C be the event of choosing a boy in 3rd trial
By the multipication theorem,

$$P(A\cap B\cap C)=P(A)P\left(\frac{B}{A}\right)P\left(\frac{C}{A\cap B}\right)$$

$$=\frac{12}{16}\times\frac{11}{15}\times\frac{10}{14}=\frac{11}{28}$$

- **Independent Events :** Two events A and B of an experiment are said to be independent if occurence of A cannot influence the happening of the event B

If A & B are independent then $P(B/A)=P(B)$ & $P(A/B)=P(A)$.

- Two events A and B are independent if and only if $P(A\cap B)=P(A).P(B)$
- If A, B and C are independent, then $P(A\cap B\cap C)=P(A).P(B).P(C)$
- If A, B, C and D are independent, then $P(A\cap B\cap C\cap D)=P(A).P(B).P(C).P(D)$

Note: If A & B are independent,

$P(A\cap B)=P(A)P(B)$ then

$P(A\cup B)=P(A)+P(B)-P(A).P(B)$. (or)

$P(A\cup B)=1-P(\bar{A}).P(\bar{B})$.

If A, B & C are three independent events, then $P(A\cup B\cup C)=1-P(\bar{A}).P(\bar{B}).P(\bar{C})$

- If two events A & B are independent
  a) A & $\bar{B}$ are independent.
  b) $\bar{A}$ & B are independent.
  c) $\bar{A}$ & $\bar{B}$ are independent.
- Three events A, B & C are said to be mutually independent if
  a) $P(A\cap B)=P(A).P(B)$
  b) $P(A\cap C)=P(A).P(C)$
  c) $P(B\cap C)=P(B).P(C)$
  d) $P(A\cap B\cap C)=P(A).P(B).P(C)$

  Suppose the 1st three conditions (i), (ii) & (iii) are satisfied and the last i.e., (iv) condition is not satisfied, then the three events A, B and C are said to be pair-wise independent.
- For any two independent events A and B

  $$P\{(A\cup B)\cap(\bar{A}\cap\bar{B})\}\leq\frac{1}{4}$$
- If A and B are two events, then

  max $\{0,1,...,P(\bar{A})-P(\bar{B})\}\leq P(A\cap B)$

  $\leq$ Min { P(A), P(B) }

W.E.10:- ***Let E,F and G be pairwise independent events with $P(G)>0$ and $P(E\cap F\cap G)=0$. then $P\left(\frac{\bar{E}\cap\bar{F}}{G}\right)=$ (JEE-2007)***

Sol: $P\left(\frac{\bar{E}\cap\bar{F}}{G}\right)=\frac{P\left(G\cap\left(\bar{E}\cap\bar{F}\right)\right)}{P(G)}$

$=\frac{P\left(G\cap\left(\overline{E\cup F}\right)\right)}{P(G)}=\frac{P(G)-P\left[G\cap(E\cup F)\right]}{P(G)}$

$=\frac{P(G)-P\left[(G\cap E)\cup(G\cap F)\right]}{P(G)}$

$=\frac{P(G)-P\left[(G\cap E)+(G\cap F)\right]}{P(G)}$

$=\frac{P(G)-P(G).P(E)-P(G)P(F)}{P(G)}$

$=1-P(E)-P(F)\quad=P(\bar{E})-P(F)$

➤ **Law of Total Probability :** Let in a random experiment S is a sample space and $E_1,E_2,E_3,....,E_n$ are mutually exclusive and exhaustive events. If A is any event which occur with $E_1\ or\ E_2\ or\ E_3\ or....or\ E_n$, then

$$P(A)=P(E_1).P\left(\frac{A}{E_1}\right)+P(E_2)P\left(\frac{A}{E_2}\right)+...$$
$$.....+P(E_n)P\left(\frac{A}{E_n}\right)$$

W.E.11:- ***An urn A contains 3 white and 5 black balls. Another urn B contains 6 white and 8 black balls. A ball is picked from A at random and then transferred to B. Then a ball is picked at random from B. The probability that it is a white ball is (EAMCET 2010)***

Sol: Let A_1, A_2 be the events of drawing white, black balls respectively from bag A then

$P(A_1)=3/8, P(A_2)=5/8$

Let E be the event of drawing a white ball from bag B after a ball is transfered from bag A to B.

Then $P(E/A_1)=7/15, P(E/A_2)=6/15$

$\therefore P(E)=P(A_1)P(E/A_1)+P(A_2)P(E/A_2)$

$=\frac{3}{8}\times\frac{7}{15}+\frac{5}{8}\times\frac{6}{15}=\frac{17}{40}$

➤ **Baye's Theorem:** Suppose $E_1,E_2,E_3,......,E_n$ are 'n' mutually exclusive and exhaustive events in sample space S with $P(E_i)>0$ for $i=1,2,3,.....,n$ in a random experiment. Then for any event A of the random experiment,

$$P\left({E_k}/{A}\right)=\frac{P(E_k).P\left({A}/{E_k}\right)}{\sum_{i=1}^{n}P(E_i)P\left({A}/{E_i}\right)}\text{ for i=1,2,3,...,n}$$

W.E.12:- ***A man is known to speak the truth 2 out of 3 times. He throws a die and reports that it is a six. Then the probability that it is actually a six is***

Sol: Let A_1 be the event that six occurs on the die and A_2 be the event that a digit other than six occurs.

Let E be the event that the man makes a statement that six is obtained.

$P(E/A_1)=$ prob. that man speaks truth= 2/3

$P(E/A_2)=$ probability that the man does not speak truth = 1/3

By Baye's theorem,

$$P(A_1/E)=\frac{P(A_1)P(E/A_1)}{P(A_1)P(E/A_1)+P(A_2)P(E/A_2)}$$

$$=\frac{\frac{2}{3}.\frac{1}{6}}{\frac{2}{3}.\frac{1}{6}+\frac{1}{3}.\frac{5}{6}}=\frac{2}{7}$$

➤ **Geometric Probability :**

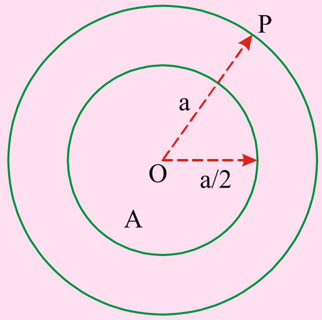
Classical definition of probability fails if the total number of outcomes of an experiment is infinite. Then the probability that a point selected in a given region will be in a specified part of it is called geometrical probability or probability in continuous. Thus the probability P is given by

$$P=\frac{\text{Measure of specified part of the region}}{\text{Measure of the whole region}}$$

Where measure refers to the length, area, volume of the region when we are dealing with one, two or three dimensional space respectively.

W.E.13:- ***A point is selected at random inside a circle of radius 'a' then the probability that it is nearer to the centre than the circumference is***

Sol:



Let O be the centre and P be any interior point. The distance OP varies from O to A. All points which lie on or inside a concentric circle with radius a/2 will be closer to the centre, than the circumference. Total area = Area of the bigger circle with radius a = πa^2.

Favourable area n(A)= $\pi\left(a/2\right)^2=\frac{1}{4}\pi a^2$

$\therefore$ Probability of A = P(A) = $\frac{\frac{1}{4}\pi a^2}{\pi a^2}=\frac{1}{4}$

NOTE:

➢ When 'n' fair coins are tossed the probability of getting exactly 'r' $(\le n)$ heads (or tails) $=\frac{{}^nc_r}{2^n}$

➢ When 'n' fair coins are tossed the probability of getting atleast one head (or tail) $=1-\frac{1}{2^n}$

➢ A coin is tossed (m+n) times (m>n), then the probability of getting at least 'm' consecutive heads is $\frac{n+2}{2^{m+1}}$

W.E.14:- ***A coin is tossed 7 times then the probability of getting atleast 4 consecutive heads is***

Sol: Here n = 3, m = 4

Required probability $=\frac{n+2}{2^{m+1}}=\frac{5}{32}$

➢ A coin is tossed (m+n) times (m>n), then the probability of getting exactly 'm' consecutive heads is $\frac{n+2}{2^{m+2}}$

W.E.15:- ***A coin is tossed 10 times, then the probability of getting exactly 6 consecutive heads is***

Sol: Here n = 10-6 = 4 ; m = 6

Required probability $=\frac{n+2}{2^{m+2}}=\frac{6}{2^8}=\frac{3}{128}$

➢ When 2 fair dice are rolled, the number of favourable cases to get sum 'k' is given by

$$\begin{cases} k-1 & for\ 2\le k\le 7 \\ 13-k & for\ 8\le k\le 12 \end{cases}$$

W.E.16:- ***Two fair dice are rolled. The probability of the sum of digits on their faces to be greater than or equal to 10 is [EAMCET 2013]***

Sol: n(A)=3+2+1=6, n(S)= 6^2 ; $\therefore$ p(A)= $\frac{6}{6^2}=\frac{1}{6}$

➢ When 3 fair dice are rolled, the number of favourable cases to get sum 'k' is given by

$$\begin{cases} \frac{1}{2}(k-1)(k-2) & for\ 3\le k\le 8 \\ 25 & for\ k=9,12 \\ 27 & for\ k=10,11 \\ \frac{1}{2}(20-k)(19-k) & for\ 13\le k\le 18 \end{cases}$$

W.E.17:- ***If 3 fair dice are rolled once, the probability of getting the sum atleast 10 is***

Sol: n(A)=27+27+25+21+15+10+6+3+1=135

n(S)= 6^3 ; p(A)= $\frac{135}{6^3}=\frac{5}{8}$

➢ When r fair dice are rolled, the number of favourable cases to get sum 'k'

= coefficient of x^k in $(x^1+x^2+.......+x^6)^r$

➢ If r squares are selected at random from a chess board the probability that they lies on a diagonal line is

$$=\frac{4[{}^7C_r+..........+{}^rC_r]+2({}^8C_r)}{{}^{64}C_r}\quad (1\le r\le 7)$$

$$=\frac{2}{{}^{64}C_8}\quad \text{if } r=8$$

W.E.18:- ***If three squares are choosen at random on a chess board. The probability that they lie on a diagonal line is***

Sol : $n(S)={}^{64}C_3$

$n(E)=4\left({}^3C_3+{}^4C_3+{}^5C_3+{}^6C_3+{}^7C_3\right)+2\left({}^8C_3\right)$

$=952 \qquad \therefore P(E)=\frac{952}{{}^{64}C_3}$

➢ **Two persons game :** If 'p' and 'q' are the probability of success, failure of a game in which A and B play and if A starts the game then,

i) probability of A's win $=\frac{p}{1-q^2}=\frac{1}{1+q}$

ii) probability of B's win $=\frac{qp}{1-q^2}=\frac{q}{1+q}$

➢ **Three persons game :** If 'p' and 'q' are the probability of success, failure of a game in which A, B and C play in order if A starts the game then,

i) probability of A's win $=\frac{p}{1-q^3}=\frac{1}{1+q+q^2}$

ii) probability of B's win $=\frac{qp}{1-q^3}=\frac{q}{1+q+q^2}$

iii) probability of C's win $=\frac{q^2p}{1-q^3}=\frac{q^2}{1+q+q^2}$

iv) the ratio of their success = 1 : q : q^2

W.E.19:- ***Three persons A,B and C toss a coin. The person who first throws a head wins. If A starts the game what are their respective probabilities of their winning.***

Sol: Let p = probability of getting a head $=\frac{1}{2}$

q = 1-p=$\frac{1}{2}$;

p(A)=$\frac{p}{1-q^3}=\frac{\frac{1}{2}}{1-1/8}=\frac{4}{7}$

p(B)=$\frac{qp}{1-q^3}=\frac{2}{7}$; p(C)=$\frac{q^2p}{1-q^3}=\frac{1}{7}$

➢ **Probability regarding 'n' letters and 'n' envelopes :** If 'n' letters are put at random in the 'n' addressed envelopes, the probability that

i) All the letters are in right envelopes is $\frac{1}{n!}$

ii) Exactly one letter in wrong envelope is 0

iii) At least one letter may be in wrongly addressed envelope is $1-\frac{1}{n!}$

iv) Exactly 'r' $(r\neq 1)$ letters are in wrong envelopes is

$\frac{{}^nP_r}{n!}\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+......+\frac{(-1)^r}{r!}\right)$

v) All the letters may be in wrong envelopes is $\frac{{}^nP_n}{n!}\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+......+\frac{(-1)^n}{n!}\right)$

W.E.20:- ***3 letters are to be placed in 3 addressed envelopes. If the letters are placed at random. The probability that***

Sol: i) All the letters are in right envelopes is $\frac{1}{3!}=\frac{1}{6}$

ii) Exactly one letter will be placed in wrong envelopes is $\frac{3p_1}{3!}\left(1-\frac{1}{1!}\right)=0$

iii) At least one letter may be in wrongly addressed envelopes is $1-\frac{1}{3!}=\frac{5}{6}$

iv) Exactly two letters will be placed in wrong addressed envelope is $\frac{3p_2}{3!}\left(1-\frac{1}{1!}+\frac{1}{2!}\right)=\frac{1}{2}$

v) All the letters will be placed in wrongly addressed envelopes is $\frac{3p_3}{3!}\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}\right)=\frac{1}{2}-\frac{1}{6}=\frac{1}{3}$

➢ Suppose a word is given. In that let the number of vowels be V, number of consonants be C and the total number of letters be T. If the letters of the word are arranged at random then the probability that

i) Relative positions of vowels and consonants do not change $=\frac{V!C!}{T!}$

ii) The order of the vowels do not change

$$= \frac{1}{\text{no. of ways of arranging the vowels}}$$

iii) The order of the vowels as well as consonants do not chanage

$$= \frac{1}{\text{(no. of ways of arranging vowels)(no. of ways of arranging consonants)}}$$

iv) The order of the consonants do not change

$$= \frac{1}{\text{no. of ways of arranging consonants}}$$

W.E.21:- ***If the letters of word 'PROBABILITY' are arranged at random. The probability that 1) relative position of vowels and consonants reamins unaltered. 2) the order of vowels remains the same. 3) the order of vowels and consonants remains the same.***

Sol: The number of letters of 'PROBABILITY' is 11 of which vowels are 4. (A,I,I,O) and consonants are 7(B,B,L,P,R,T,Y).

The number of arrangements of vowels $= \frac{4!}{2!}$

The number of arrangements of consonants $= \frac{7!}{2!}$

1. Req. probability $= \frac{(4!/2!).(7!/2!)}{11!/2!.2!} = \frac{4!7!}{11!}$

2. Required probability $= 1/\frac{4!}{2!} = 2!/4! = 1/12$

3. Req. probability $= \frac{1}{4!/2!\ \ 7!/2!} = 2!2!/4!7!$

➢ Out of 'n' pairs of shoes if r(<n) shoes are selected at random then the probability that

i) there is no pair $= \frac{{}^nC_r.2^r}{{}^{2n}C_r}$

ii) there is atleast one pair $= 1 - \frac{{}^nC_r.2^r}{{}^{2n}C_r}$

iii) there are exactly k pairs $= \frac{{}^nC_k.{}^{n-k}C_{r-2k}2^{r-2k}}{{}^{2n}C_r}$

W.E.22:- From a heap containing 10 pairs of shoes, 6 shoes are selected at random. Then the probability that there is
i) no pair in the selected shoes
ii) atleast one pair in the selected shoes
iii) 2 pairs in the selected shoes

Sol: $n(S) = {}^{20}C_6$

i) $P(A) = \frac{{}^{10}C_6 \times 2^6}{{}^{20}C_6}$

ii) P(atleast one pair in the selected 6shoes) = 1-P(no pair in the selected 6 shoes)

$$= 1 - \frac{{}^{10}C_6 \times 2^6}{{}^{20}C_6}$$

iii) $P(A) = \frac{{}^{10}C_2 \times {}^8C_2 \times 2^2}{{}^{20}C_6} = \frac{42}{323}$

➢ Let A be a set containing 'n' elements. Then the probability that 'r' subsets of A selected at random, have no element in common is $\left(\frac{2^r - 1}{2^r}\right)^n$

W.E.23:- ***If three subsets of a set containing 5 elements are selected at random. The probability that they have no element in common is ....***

Sol: Here n = 5 , r = 3

required probability $= \left(\frac{2^3 - 1}{2^3}\right)^5 = \left(\frac{7}{8}\right)^5$

➢ A is a set containing n elements. A subset P of A is chosen at random. The set A is reconstructed by replacing the elements of P, a subset Q of A is again chosen at random. The probability that

i) $P \cap Q$ contains exactly r elements is $\frac{{}^nC_r 3^{n-r}}{4^n}$

ii) $P \cup Q$ = A is $\frac{3^n}{4^n}$

iii) $P \cup Q$ and $P \cap Q = \phi$ is $\frac{2^n}{4^n} = \left(\frac{1}{2}\right)^n$

iv) Q is a subset of P is $\frac{3^n}{4^n} = \left(\frac{3}{4}\right)^n$

W.E.24:- ***A is a set containing n elements. A subset P of A is chosen at random. The set A is reconstructed by replacing the elements of the subset of P. A subset Q of A is chosen at random. The probability that P and Q have no common elements is***

Sol: Let A$=\{a_1,a_2,....,a_n\}$ for each $a_i\,(1\le i\le n)$ we have the following four choices.

i) $a_i\in P, a_i\in Q$ ii) $a_i\in P, a_i\notin Q$

iii) $a_i\notin P, a_i\in Q$ iv) $a_i\notin P, a_i\notin Q$

Thus the total number of ways of choosing P and Q is 4^n.

Out of these i) is not favourable and hence the number of favourable ways is 3^n

$\therefore$ Prob. of the req. event $=3^n/4^n=(3/4)^n$.

➢ If m things are distributed among M persons, then the probability that a particular person get r things (r < m) is $\dfrac{m\,c_r\times(M-1)^{M-r}}{M^m}$

W.E.25:- ***5 biscuits are to be distributed among 10 beggars. What is the probability that a particular beggar receives 3 biscuits ?***

Sol: $P(E)=\dfrac{{}^5C_3\,(10-1)^{5-3}}{10^5}=\dfrac{10\times9^2}{10^5}=\dfrac{81}{10000}$

W.E.26:- ***From (2n+1) consecutive positive integers 3 are selected at random. The probability that they are in A.P.=*** $\dfrac{3n}{4n^2-1}$

Sol: We can choose 3 numbers out of $(2n+1)$ in $(2n+1)_{C_3}$ ways. Let the 3 numbers be x_1,x_2,x_3.

Now x_1,x_2,x_3 are in A.P $\Rightarrow x_1+x_3=2x_2$

$\Rightarrow x_1,x_3$ are odd or x_1,x_3 are even

x_1,x_3 can be choosen in $(n+1)_{C_2}+n_{C_2}=n^2$ ways

Required probability $=\dfrac{n^2}{(2n+1)_{C_3}}=\dfrac{3n}{4n^2-1}$

W.E.27:- ***From 2n consecutive positive integers 3 are selected at random. The probability that the numbers on them are in A.P is*** $\dfrac{3}{2(2n-1)}$

Sol: We can choose 3 numbers out of 2n in $(2n)_{C_3}$ ways

Let the 3 numbers be x_1,x_2,x_3

Now x_1,x_2,x_3 are in A.P $\Rightarrow x_1+x_3=2x_2$

$\Rightarrow x_1,x_3$ are odd or x_1,x_3 are even

x_1,x_3 can be choosen in ${}^nC_2+{}^nC_2=n(n-1)$ ways.

Required probability $=\dfrac{n(n-1)}{(2n)_{C_3}}=\dfrac{3}{2(2n-1)}$

➢ If n men among whom A and B sit along a circle then the odds aganist A and B sitting together $=\dfrac{n-3}{n-1}:\dfrac{2}{n-1}=n-3:2$

W.E.28:- ***If 10 persons are to sit round a table. The odds aganist two specified persons together are.***

Sol: Required ratio = n - 3 : 2 = 10-3 : 2 = 7 : 2

➢ A man has a bunch of n keys of which only one fits the door. He tries them successively without repetition. Then, the probability that the door is opened at any trail (less than n)= $\dfrac{1}{n}$

W.E.29:- ***The key for opening a door is in a bunch of 10 keys. A man attemps to open the door by trying the keys at random discording the wrong key. The probabiltiy that the door is opened in the 5th trail is***

Sol: Required probability = $\dfrac{1}{10}$

➢ If 'n' men among whom A and B sit along a circle then the probability that there will be exactly 'r' persons between A and B = $\dfrac{2}{n-1}$ (in both directions)

W.E.30:- ***A and B stand in a ring with 10 other persons. If the arrangement of the 12 persons is at random. What is the probability that there are exactly 3 persons between A and B.***

Sol: Here n = 12, r = 3

Required probability $= \dfrac{2}{12-1} = \dfrac{2}{11}$

➢ Using the vertices of a polygon having n sides a triangle is constructed at random. The probability that the triangle so formed is such that no side of the polygon is side of the triangle is $\dfrac{(n-4)(n-5)}{(n-1)(n-2)}$

➢ Out of n persons sitting at a round table, three persons are selected at random then the probability that no two of them are consecutive is $\dfrac{(n-4)(n-5)}{(n-1)(n-2)}$

W.E.31:- ***If 100 people sit round a table, the probability of selecting 3 people so that no two of them are next to each other is***

Sol: Required probability =

$$\frac{(100-4)(100-5)}{(100-1)(100-2)} = \frac{96\times 95}{99\times 98}$$

➢ The probability that two friends will be able to meet each other between 'a' hours and 'b' hours, not waiting for each other more than 'c' hours is

$$\frac{(b-a)^2-(1-c)^2}{(b-a)^2}$$

W.E.32:- ***Two friends agreed to meet on next day at some place between 2pm and 3pm with understanding that none waits for the other more than 10 minutes. The probability that they can meet is***

Sol: Here a = 2,
b = 3,
c=10 min = 1/6 hr.

Required probability =

$$\frac{(3-2)^2-\left(1-\frac{1}{6}\right)^2}{(3-2)^2} = 11/36$$

➢ If n whole numbers are taken at random and multiplied together then the probability that the last digit of the product is

i) either 1,3,7 or 9 is $\dfrac{4^n}{10^n} = (2/5)^n$

ii) either 2,4,6 or 8 is $\dfrac{8^n-4^n}{10^n} = \dfrac{4^n-2^n}{5^n}$

iii) 5 is $\dfrac{5^n-4^n}{10^n}$

iv) 0 is $\dfrac{10^n-8^n-5^n+4^n}{10^n}$

➢ Let S be a finite set containing 'n' elements.Then

i) The total number of binary operations on 'S' is equal to n^{n^2}

ii) The total number of commutative binary operations on 'S' is equal to $n^{\frac{n(n+1)}{2}}$

iii) The probability of selecting a cumulative binary operation from a set S to itself $= \dfrac{n^{\frac{n(n+1)}{2}}}{n^{n^2}}$

iv) The probability of selecting an injective mapping from a set 'S' to itself is $\dfrac{n!}{n^n}$

v) The probability of selecting a bijection mapping from a set 'S' to itself is $\dfrac{n!}{n^n}$

W.E.33:- ***A binary operation is chosen at random from the set of binary operations on a set A consisting 12 elements. The probability that the binary operation is commutative is***

Sol: Total number of binary operations on $A = 12^{12^2}$ and number of commutative binary operations on A is $12^{12\left(\frac{12+1}{2}\right)}$

Required probability $= \dfrac{12^{12\left(\frac{12+1}{2}\right)}}{12^{12^2}} = \dfrac{1}{12^{66}}$

C.U.Q.

1. The probability of an impossible event is

1) $\frac{1}{2}$ 2) 1 3) 0 4) $\frac{1}{4}$

2. The probability of obtaining exactly 'r' heads and $(n-r)$ tails, when we toss n unbiased coins is

1) $\frac{r}{n}$ 2) $\frac{(n-r)}{n}$ 3) $\frac{{}^nc_r}{2^n}$ 4) $\frac{{}^nc_r}{3^n}$

3. An unbiased coin is tossed n times. The probability that head will present itself, odd number of times is

1) $\frac{1}{4}$ 2) $\frac{1}{3}$ 3) $\frac{1}{2}$ 3) $\frac{1}{5}$

4. If A and B are any two events in a sample space S then $P(A\cup B)$ is

1) $\geq P(A)+P(B)$ 2) P(A) + P(B)

3) $\leq P(A)+P(B)$ 4) $P(A\cap B)$

5. If $A\subset B$ then $P(A\cap B^C)=$

1) 1 2) 0 3) P(A) 4) P(B)

6. If A and B are two mutually exclusive events, then the relation between $P(\bar{A})$ and $P(B)$ is

1) $P(B)\geq P(\bar{A})$ 2) $P(B)\leq P(\bar{A})$

3) $P(B)=P(A)$ 4) $P(B)<P(A)$

7. If E_1,E_2 are two events with $E_1\cap E_2=\varnothing$ then $P(\bar{E}_1\cap\bar{E}_2)=$

1) $P(E_1)+P(E_2)$ 2) $P(\bar{E}_1)-P(E_2)$

3) $P(\bar{E}_1)+P(E_2)$ 4) $P(E_1)-P(E_2)$

8. If A & B are two events then $P\{(A\cap\bar{B})\cup(\bar{A}\cap B)\}=$

1) $P(A\cup B)-P(A\cap B)$ 2) $P(A\cup B)+P(A\cap B)$

3) $P(A)+P(B)$ 4) $P(A)+P(B)+P(A\cap B)$

9. If $P\left(\frac{A}{C}\right)>P\left(\frac{B}{C}\right)$ and $P\left(\frac{A}{\bar{C}}\right)>P\left(\frac{B}{\bar{C}}\right)$ then the relation between P(A) and P(B) is

1) P(A)=P(B) 2) $P(A)\leq P(B)$

3) $P(A)>P(B)$ 4) $P(A)\geq P(B)$

10. If $P\left(\frac{B}{A}\right)<P(B)$, the relation between $P\left(\frac{A}{B}\right)$ and P(A) is

1) $P\left(\frac{A}{B}\right)<\frac{1}{2}P(A)$ 2) $P\left(\frac{A}{B}\right)>P(A)$

3) $P\left(\frac{A}{B}\right)<P(A)$ 4) $P\left(\frac{A}{B}\right)<2P(A)$

11. If C and D are two events such that $C\subset D$ and P(D) $\neq 0$, then the correct statement among the following is [AIEEE 2011]

1) $P(C/D)<P(C)$ 2) $P(C/D)=\frac{P(D)}{P(C)}$

3) $P(C/D)=P(C)$ 4) $P(C/D)\geq P(C)$

12. If A is an independent event to itself then P(A)=

1) 0 2) 1 3) 0,1 4) $\frac{1}{2}$

13. If $A_1,A_2,A_3.....,A_n$ are n independent events such that $P(A_k)=\frac{1}{k+1},K=1,2,.....,n;$ then the probability that none of the n events occur is [EAMCET 2010]

1) $\frac{1}{n+1}$ 2) $\frac{n}{n+1}$ 3) $\frac{n}{(n+1)(n+2)}$ 4) $\frac{1}{(n+1)!}$

14. A is a set containing *'n'* elements. A subset P of A is chosen at random. The set A is reconstructed by replacing the elements of the subset of P, a subset Q of A is again chosen at random. The probability that $P\cup Q=A$ and $P\cap Q=\phi$ is

1) $\left(\frac{3}{4}\right)^n$ 2) $\left(\frac{1}{4}\right)^n$ 3) $\left(\frac{1}{2}\right)^n$ 4) $\left(\frac{1}{3}\right)^n$

15. A is a set containing n elements. A subset P of A is chosen at random. The set A is reconstructed by replacing the elements of the subset of P. A subset Q of A is chosen at random. The probability that P and Q have no common element is

1) $\frac{2^n}{3^n}$ 2) $\frac{2^n}{4^n}$ 3) $\frac{3^n}{4^n}$ 4) $\frac{3^n}{5^n}$

16. An urn contains *'w'* white balls and *'b'* black balls. Two players Q and R alternatively draw a ball with replacement from the urn. The player who draws a white ball first wins the game. If Q begins the game, The probability of his winning the game is

1) $\frac{w}{w+b}$ 2) $\frac{w+b}{w+3b}$ 3) $\frac{w+b}{w+2b}$ 4) $\frac{w+b}{w^2+b}$

C.U.Q. - KEY

1)3	2)3	3) 3	4) 3	5)2	6) 2
7) 2	8) 1	9) 3	10) 3	11) 4	12) 3
13) 1	14) 3	15) 3	16) 3		

C.U.Q. - HINTS

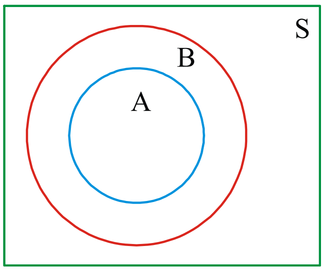
1.. n (A) = **0**

2.. From n coins getting r heads $\Rightarrow$ getting $(n-r)$ tails. $\therefore n(E) = Getting\ r\ heads = n_{C_r}$ $n(S) = 2^n$

3. $P(E) = \dfrac{2^{n-1}}{2^n} = \dfrac{1}{2}$

4. $P(A \cup B) = P(A) + P(B) - P(A \cap B) \le P(A) + P(B)$

5.



$A \subset B \Rightarrow A \cup B^C = \phi \quad \Rightarrow P\left(A \cap B^C\right) = 0$

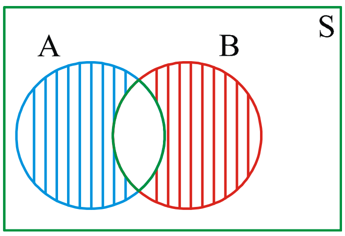
6. $P(A) + P(B) \le 1$; $P(B) \le P(\overline{A})$

7. L.H.S. = $P(\overline{E}_1 \cap \overline{E}_2) = 1 - P(E_1 \cup E_2)$

$= 1 - [P(E_1) + P(E_2)] \quad \because E_1 \cap E_2 = \phi$

$= [1 - P(E_1)] - P(E_2) = P(\overline{E}_1) - P(E_2)$

8.



$P\{(A \cap \overline{B}) \cup (\overline{A} \cap B)\} = P(A \cap \overline{B}) + P(\overline{A} \cap B) =$

$P(A) - (A \cap B) + P(B) - P(A \cap B)$

$= P(A \cup B) - P(A \cap B)$

9. $P\left(\dfrac{A}{C}\right) > P\left(\dfrac{B}{C}\right) \Rightarrow P(A \cap C) > P(B \cap C)$

$\Rightarrow P(A \cap C) - P(B \cap C) > 0$

$P\left(\dfrac{A}{\overline{C}}\right) > P\left(\dfrac{B}{\overline{C}}\right) \Rightarrow P(A) - P(A \cap C) > P(B) - P(B \cap C)$

$P(A) - P(B) > P(A \cap C) - P(B \cap C) > 0$

10. Given $P\left(\dfrac{B}{A}\right) < P(B) \Rightarrow P\left(\dfrac{A}{B}\right) < P(A)$

$\dfrac{P(A \cap B)}{P(A)} < P(B) \Rightarrow \dfrac{P(A \cap B)}{P(B)} < P(A)$

11. $P(C/D) = \dfrac{P(C \cap D)}{P(D)} = \dfrac{P(C)}{P(D)} \ge P(C)$

12. $P(A \cap A) = P(A)P(A) \Rightarrow P(A) = \{P(A)\}^2$

13. given $P(A_1) = \dfrac{1}{2}, P(A_2) = \dfrac{1}{3} .... P(A_n) = \dfrac{1}{n+1}$

$P\left(\overline{A}_1 \cap \overline{A}_2 \cap ..... \cap \overline{A}_n\right) = \dfrac{1}{2} \times \dfrac{2}{3} \times \dfrac{3}{4} \times .... \times \dfrac{n}{n+1} = \dfrac{1}{n+1}$

14. Let $A = \{a_1, a_2, a_3, ..., a_n\}$

for each $a_k\ (1 \le k \le n)$ we have the following 4 choices

a) $a_k \in P$ and $a_k \in Q$ b) $a_k \in P$ and $a_k \notin Q$

c) $a_k \notin P$ and $a_k \in Q$ d) $a_k \notin P$ and $a_k \notin Q$

The total number of ways choosing the subsets P and Q is 4^n.

Out of the 4 choices (a) and (d) are not favourables for the occurrance of the event $P \cup Q = A$ and $P \cap Q = \phi$

The number of favourables is 2^n.

Required Probability $= \left(\dfrac{2}{4}\right)^n = \left(\dfrac{1}{2}\right)^n$

15. $n(S) = 2^n \times 2^n = 4^n$; $n(E) = 3^n$; $P(E) = \left(\dfrac{3}{4}\right)^n$

16. Let p be the probability of drawing a white ball from bag by any person in one trail.

Here $p = \dfrac{w}{w+b}$; Let $q = 1 - p = 1 - \dfrac{w}{w+b} = \dfrac{b}{w+b}$;

$P(Q \text{ wins}) = p + q^2p + q^4p + ... \infty$

$= \dfrac{p}{1-q^2} = \dfrac{\dfrac{w}{w+b}}{1 - \dfrac{b^2}{(w+b)^2}}$

LEVEL-I (C.W)

PROBLEMS BASED ON CLASSICAL DEFINITION OF PROBABILITY

1. If A and B are two Mutually Exclusive events in a sample space S such that P(B) = 2 P(A) and $A \cup B = S$ then P(A)=

1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{1}{4}$ 4) $\frac{1}{5}$

2. From the set of numbers {1, 2, 3, 4, 5, 6, 7, 8} two numbers are selected at random without replacement. The probability that their sum is more than 13 is

1) $\frac{1}{14}$ 2) $\frac{2}{7}$ 3) $\frac{3}{7}$ 4) $\frac{4}{7}$

3. In a room there are 6 couples. Out of them if 4 are selected at random, the probability that they may be couples is

1) $\frac{4}{33}$ 2) $\frac{2}{33}$ 3) $\frac{1}{33}$ 4) $\frac{3}{33}$

4. Two letters are taken at random from the word HOME. The probability that at least one is a vowel is

1) $\frac{1}{2}$ 2) $\frac{2}{3}$ 3) $\frac{5}{6}$ 4) $\frac{1}{4}$

5. Three electric lamps are to be fitted in a room. Three bulbs are chosen at random from 10 bulbs having 6 good bulbs and fitted. The chance that the room is lighted is

1) $\frac{1}{30}$ 2) $\frac{2}{30}$ 3) $\frac{28}{30}$ 4) $\frac{29}{30}$

6. 5 persons entered the lift cabin on the ground floor of an eight floor house. Suppose each of them independently leave the cabin at any floor beginning with 1st floor, the probability of all the 5 persons leaving at different floor is

1) $\frac{{}^8P_5}{8^5}$ 2) $\frac{{}^7P_5}{7^5}$ 3) $\frac{5}{8}$ 4) $\frac{1}{8}$

7. Four numbers are chosen at random from $\{1,2,3,......,40\}$. The probability that they are not consecutive is

1) $\frac{1}{2470}$ 2) $\frac{4}{7969}$ 3) $\frac{2469}{2470}$ 4) $\frac{7965}{7969}$

8. Two numbers are chosen at random from {1, 2, 3, 4, 5, 6, 7, 8} at a time. The probability that smaller of the two numbers is less than 4 is [EAMCET 2013]

1) $\frac{7}{14}$ 2) $\frac{8}{14}$ 3) $\frac{9}{14}$ 4) $\frac{10}{14}$

9. Two numbers are selected at random from 1, 2, 3, ... 100 without replacement. The probability that the minimum of the two numbers is less than 70 is

1) $\frac{30_{C_2}}{100_{C_2}}$ 2) $1-\frac{30_{C_2}}{100_{C_2}}$ 3) $\frac{31_{C_2}}{100_{C_2}}$ 4) $1-\frac{31_{C_2}}{100_{C_2}}$

10. Two symmetrical dice are thrown at a time. If the sum of points on them is 7, the chance that one of them will show a face with 2 points is

1) $\frac{1}{8}$ 2) $\frac{1}{3}$ 3) $\frac{2}{3}$ 4) $\frac{2}{8}$

11. Three students A, B and C are to take part in a swimming competition. The probability of A's winning or B's winning each is 3 times the probability of C's winning. The probability of C's winning if there is no tie is

1) $\frac{4}{7}$ 2) $\frac{3}{7}$ 3) $\frac{2}{7}$ 4) $\frac{1}{7}$

12. Let A be a set of 4 elements. From the set of all functions from A to A, the probability that it is an into function is

1) $\frac{3}{32}$ 2) 0 3) $\frac{29}{32}$ 4) 1

13. S = {1, 2, 3, ... 20} if 3 numbers are chosen at random from S, the probability for they are in A.P. is

1) $\frac{3}{38}$ 2) $\frac{35}{33}$ 3) $\frac{33}{35}$ 4) $\frac{1}{38}$

14. The probability of choosing randomly a number c form the set {1, 2, 3, ...9} such that the quadratic equation $x^2+4x+c=0$ has real roots is : [EAMCET 2009]

1) $\frac{1}{9}$ 2) $\frac{2}{9}$ 3) $\frac{3}{9}$ 4) $\frac{4}{9}$

15. A class has fifteen boys and five girls. Suppose three students are selected at random form the class. The probability that there are two boys and one girl is : [EAMCET 2011]

1) $\frac{35}{76}$ 2) $\frac{35}{38}$ 3) $\frac{7}{76}$ 4) $\frac{35}{72}$

16. The probability of getting at least one head when we toss 3 unbiased coins is

1) $\frac{3}{8}$ 2) $\frac{5}{8}$ 3) $\frac{7}{8}$ 4) $\frac{1}{8}$

17. A tosses 2 coines while B tosses 3. The probability that B obtains more number of heads is

1) $\frac{1}{4}$ 2) $\frac{1}{3}$ 3) $\frac{1}{2}$ 4) $\frac{3}{4}$

18. The probability of getting at least 2 heads, when an unbiased coin is tossed 6 times is

1) $\frac{63}{64}$ 2) $\frac{57}{64}$ 3) $\frac{7}{64}$ 4) $\frac{37}{64}$

19. A fair coin is tossed 4 times. The probability that heads exceed tails in number is

1) $\frac{3}{16}$ 2) $\frac{1}{4}$ 3) $\frac{5}{16}$ 4) $\frac{7}{16}$

20. An unbiased coin is tossed to get 2 points for turning up a head and one point for the tail. If three unbiased coins are tossed simultaneously, then the prob. of getting a total of odd no. of points is

1) 1/2 2) 1/4 3) 1/8 4) 3/8

21. Two symmetrical dice are thrown. The probability of getting a sum of 6 points is

1) $\frac{4}{36}$ 2) $\frac{5}{36}$ 3) $\frac{6}{36}$ 4) $\frac{1}{36}$

22. Two uniform dice marked 1 to 6 are thrown together. The probability that the total score on them is either minimum or maximum.

1) $\frac{4}{36}$ 2) $\frac{5}{36}$ 3) $\frac{2}{36}$ 4) $\frac{1}{36}$

23. Two uniform dice marked 1 to 6 are thrown together. The probability that the score on the two dice is at least seven is

1) $\frac{5}{12}$ 2) $\frac{7}{12}$ 3) $\frac{3}{4}$ 4) $\frac{1}{2}$

24. Two uniform dice marked 1 to 6 are thrown together. The probability that the sum is even is

1) $\frac{1}{4}$ 2) $\frac{1}{3}$ 3) $\frac{1}{2}$ 4) $\frac{1}{12}$

25. Two dice are thrown. The probability that the absolute difference of points on them is 4 is

1) $\frac{1}{7}$ 2) $\frac{1}{8}$ 3) $\frac{1}{9}$ 4) $\frac{1}{6}$

26. Three symmetrical dice are thrown. The probability of having different points on them is

1) $\frac{35}{36}$ 2) $\frac{4}{9}$ 3) $\frac{5}{9}$ 4) $\frac{1}{36}$

27. A and B throw with 3 dice. If A throws a sum of 16 points, the probability of B throwing a higher sum is

1) $\frac{7}{54}$ 2) $\frac{5}{54}$ 3) $\frac{1}{54}$ 4) $\frac{3}{54}$

28. On a symmetrical die the numbers 1, -1, 2, -2, 3 and 0 are marked on its 6 faces. If such a die is thrown 3 times, the probability that the sum of points on them is 6 is

1) $\frac{5}{27}$ 2) $\frac{5}{54}$ 3) $\frac{5}{108}$ 4) $\frac{3}{108}$

29. A symmetrical die is thrown 4 times. The probability that 3 and 6 will turn up exactly 2 times each is

1) $\frac{1}{6^4}$ 2) $\frac{1}{6^3}$ 3) $\frac{1}{6^2}$ 4) $\frac{1}{6}$

30. Two symmetrical dice are thrown. The probability of throwing a doublet such that their sum is greater than 9 is

1) $\frac{1}{36}$ 2) $\frac{1}{18}$ 3) $\frac{1}{9}$ 4) $\frac{1}{72}$

31. Two symmetrical dice are rolled. The probability that sum of the points on them is divisible by 5 is

1) $\frac{2}{9}$ 2) $\frac{4}{9}$ 3) $\frac{7}{36}$ 4) $\frac{5}{9}$

32. Let S be the sample space of the random experiment of throwing simultaneously two unbiased dice with six faces (numbered 1 to 6) and let $\mathrm{E_k} = \{(a,b) \in S : ab = k\}$ $for\ k \geq 1$

If $p_k = P(E_k)$ for $k \geq 1$ then correct among the following, is

1) $p_1 < p_{30} < p_4 < p_6$ 2) $p_{36} < p_6 < p_2 < p_4$

3) $p_1 < p_{11} < p_4 < p_6$ 4) $p_{36} < p_{11} < p_6 < p_4$

33. A coin and a six faced die, both unbiased, are thrown simultaneously. The probability of getting a head on the coin and an odd number on the die is

1) 1/2 2) 3/4 3) 1/4 4) 2/3

34. Six faces of a unbiased die are numbered with 2, 3, 5, 7, 11 and 13. If two such dice are thrown, then the probability that the sum on the uppermost faces of the dice is an odd number is

1) 5/18 2) 5/36 3) 13/18 4) 25/36

35. The probability that a leap year will have 53 Sundays and 53 Mondays is

1) $\frac{3}{7}$ 2) $\frac{2}{7}$ 3) $\frac{1}{7}$ 4) $\frac{4}{7}$

36. The probability that the February of a leap year will have 5 saturdays is

1) $\frac{3}{7}$ 2) $\frac{2}{7}$ 3) $\frac{1}{7}$ 4) $\frac{4}{7}$

37. The probability that 13th day of the randomly chosen month is a Friday is

1) $\frac{1}{7}$ 2) $\frac{1}{12}$ 3) $\frac{1}{84}$ 4) $\frac{1}{42}$

38. In a non leap year the probability of getting 53 Sundays or 53 Tuesdays or 53 Thursdays

1) $\frac{1}{7}$ 2) $\frac{2}{7}$ 3) $\frac{3}{7}$ 4) $\frac{4}{7}$

39. It is given that a leap year has 53 Sundays, the probability that it has 53 Mondays is

1) $\frac{1}{2}$ 2) $\frac{1}{7}$ 3) $\frac{6}{7}$ 4) $\frac{2}{7}$

40. A pack of cards is distributed among four hands equally. The probability that 5 spades, 3 clubs, 3 hearts and the rest diamonds may be in a particular hand is

1) $\frac{^{4}c_1 \times ^{13}c_5 \times ^{13}c_3 \times ^{13}c_2}{^{52}c_{13}}$ 2) $\frac{^{13}c_5 \times ^{26}c_6 \times ^{13}c_2}{^{52}c_{13}}$

3) $\frac{^{13}c_5 \times ^{13}c_3 \times ^{13}c_3 \times ^{13}c_2}{^{52}c_{13}}$ 4) $\frac{^{4}c_1 \times ^{13}c_3 \times ^{13}c_3 \times ^{13}c_2}{^{52}c_{13}}$

41. A card is drawn from an ordinary pack of 52 playing cards and a gambler bets it as a spade or an ace. The probability that he wins the bet is

1) $\frac{2}{13}$ 2) $\frac{3}{13}$ 3) $\frac{4}{13}$ 4) $\frac{1}{13}$

42. In shuffling a pack of cards, four cards are accidentally dropped. The probability that the cards dropped are one from each suit is

1) $\frac{^{13}c_4}{^{52}c_4}$ 2) $\frac{13^4}{^{52}c_4}$ 3) $\frac{^{13}c_4}{13^4}$ 4) $\frac{13!}{^{52}c_4}$

43. The face cards are removed from a well shuffled pack of 52 cards. Out of the remaining cards 4 are drawn at random. The probability that they belong to different suits is

1) $\frac{13^4}{^{52}c_4}$ 2) $\frac{^{13}c_4}{^{40}c_4}$ 3) $\frac{10^4}{^{40}c_4}$ 4) $\frac{13^4}{^{40}c_4}$

44. In a game of bridge, the probability of a particular player having all the 13 cards of red colour is

1) $\frac{13^4}{^{52}c_{13}}$ 2) $\frac{^{26}c_{13}}{^{52}c_{13}}$ 3) $\frac{13\times 4}{^{52}c_{13}}$ 4) $\frac{13^2}{^{52}c_{13}}$

45. From a pack of cards, 2 cards are chosen at random. The probability of the event of one card is 10 which is not hearts and another a hearts card is

1) $\frac{1}{34}$ 2) $\frac{1}{102}$ 3) $\frac{8}{663}$ 4) $\frac{33}{34}$

46. A person draws 2 cards from a well shuffled pack of cards, the cards are replaced after noting their colour. Then another person draws 2 cards after shuffling the pack. The probability that there will be exactly 1 common card is

1) $\frac{50}{663}$ 2) $\frac{25}{663}$ 3) $\frac{100}{663}$ 4) $\frac{75}{663}$

47. Seven white balls and three black balls are randomly arranged in a row. The probability that no two black balls are placed adjacently is :

1) $\frac{1}{2}$ 2) $\frac{7}{15}$ 3) $\frac{2}{15}$ 4) $\frac{1}{3}$

48. A box contains 40 balls of the same shape and weight. Among the balls 10 are white, 16 are red and the rest are black, if two balls are drawn, the probability that one is red and one is black is

1) 3/4 2) 56/195 3) 1 4) 1/4

49. An urn contains nine balls of which three are red, four are blue and two are green. Three balls are drawn at random wifthout replacement from the urn. The probability that the three balls have different colours is: [AIEEE 2010]

1) $\frac{1}{3}$ 2) $\frac{2}{7}$ 3) $\frac{1}{21}$ 4) $\frac{2}{23}$

50. If three balls drawn from a bag contains are 4 red, 3 black and 5 white balls. The probability of drawing 2 balls of the same colour and one is of different colour is

1) $\frac{5}{44}$ 2) $\frac{15}{44}$ 3) $\frac{29}{44}$ 4) $\frac{131}{195}$

51. From a bag containing 4 white and 5 black balls 3 are drawn at random. The odds against these being all black balls is

1) 5 to 42 2) 37 to 5 3) 5 to 37 4) 42 to 5

52. A box X contains 2 white and 3 black balls and another bag Y contains 4 white and 2 black balls. One bag is selected at random and a ball is drawn from it. Then the probability for the ball chosen be white is (Eamcet 2003)

1) 2/15 2) 7/15 3) 8/15 4) 14/15

53. Box A contains 3 red and 2 black balls. Box B contains 2 red and 3 black balls. One ball is drawn at random from box A and placed in box B. Then one ball is drawn at random from the box B and placed in A. The probability that the composition of balls in the two boxes remains unaltered is

1) $\frac{9}{30}$ 2) $\frac{4}{15}$ 3) $\frac{17}{30}$ 4) $\frac{16}{30}$

54. Three '1×1' squares of a chess board having 8×8 squares being chosen at random, the chance that all the three are white is

1) $\frac{^{32}C_3}{^{64}C_3}$ 2) $\frac{^{8}C_3}{^{64}C_3}$ 3) $\frac{^{16}C_3}{^{64}C_3}$ 4) $\frac{^{4}C_3}{^{64}C_3}$

55. Three '1×1' squares of a chess board having 8×8 squares are chosen at random, the chance that 2 are of one colour and 1 is of different colour is

1) $\frac{^{32}C_3}{^{64}C_3}$ 2) $\frac{^{16}C_3}{^{64}C_3}$ 3) $\frac{2\times{}^{32}C_2\times{}^{32}C_1}{^{64}C_3}$ 4) 1

PROBLEMS ON ADDITION THEOREM

56. If A and B are two events such that $P(A)=\frac{1}{4}$, $P(A\cup B)=\frac{1}{3}$ and P(B) = P, the value of P if $A\subset B$

1) $\frac{1}{12}$ 2) $\frac{3}{4}$ 3) $\frac{2}{3}$ 4) $\frac{1}{3}$

57. Suppose that A and B are two events such that $P(A\cap B)=\frac{3}{25}$ and $P(B-A)=\frac{8}{25}$. Then P(B) = [EAMCET 2010]

1) $\frac{11}{25}$ 2) $\frac{3}{11}$ 3) $\frac{1}{11}$ 4) $\frac{9}{11}$

58. Two events A and B have the probabilities 0.25 and 0.5 respectively. The probability that both A and B occur simultaneously is 0.14. The probability that neither A nor B occurs is

1) 0.39 2) 0.29 3) 0.19 4) 0.5

59. If A,B and C are mutually exclusive and exhaustive events of a random experiment such that $P(B)=\frac{3}{2}P(A)$ and $P(C)=\frac{1}{2}P(B)$ then $P(A\cup C)=$ [EAMCET 2014]

1) $\frac{3}{13}$ 2) $\frac{6}{13}$ 3) $\frac{7}{13}$ 4) $\frac{10}{13}$

60. A and B are seeking admission into IIT. If the probability for A to be selected is 0.5 and that of both to be selected is 0.3. The at most probability of B *is*

1) 0.9 2) 0.8 3) 0.6 4) 0.4

61. There are three events A, B and C one of which and only one can happen. The odds are 7 to 3 against A and 6 to 4 against B. The odds against C are

1) 3 to 4 2) 4 to 3 3) 7 to 3 4) 3 to 7

62. A number is chosen from the first 100 natural numbers. The probability that it is a multiple of 4 or 6 is

1) $\frac{25}{100}$ 2) $\frac{16}{100}$ 3) $\frac{8}{100}$ 4) $\frac{33}{100}$

63. The results of an examination in two papers A and B for 20 candidates were recorded as follows. 8 passed in paper A, 7 passed in paper B, 8 failed in both the papers A and B. If one is selected at random, the probability that the candidate has failed in A or B is

1) $\frac{15}{20}$ 2) $\frac{16}{20}$ 3) $\frac{17}{20}$ 4) $\frac{18}{20}$

64. In a class there are 10 men & 20 women. Out of them half of the number of men & half of the number of women have brown eyes. Out of them if a person is chosen at random, the chance that for the person chosen to be a man or brown eyed person is

1) $\frac{1}{3}$ 2) $\frac{2}{3}$ 3) $\frac{3}{4}$ 4) $\frac{1}{4}$

65. If the probabilities of two dogs A and B dying within 10 years are respectively p and q, then the probability that at least one of them will be alive at the end of 10 years is

1) $p+q$ 2) $1-pq$ 3) $p+q-pq$ 4) pq

66. The probability that a man A will be alive for 20 more years is $\frac{3}{5}$ and the probability that his wife will be alive for 20 more years is $\frac{2}{3}$. The probability that only one will be alive at the end of 20 years is

1) $\frac{4}{15}$ 2) $\frac{2}{15}$ 3) $\frac{7}{15}$ 4) $\frac{8}{15}$

67. A husband and wife appear in an interview for two vacancies in the same post. The probability of husbands selection is $\frac{1}{7}$ and that of wife is $\frac{1}{5}$. The probability that both of them will be selected is

1) $\frac{24}{35}$ 2) $\frac{2}{7}$ 3) $\frac{1}{35}$ 4) $\frac{2}{35}$

68. A can hit a target 3 times in 6 shots; B, 2 times in 4 shots and C, 4 times in 4 shots. All of them fire at a target independently. The probability that the target will be hit is

1) 1 2) 0.5 3) 0.25 4) 0.125

PROBLEMS ON CONDITIONAL PROBABILITY

69. If A and B are two events such that $P(A)=\frac{3}{8}$, $P(B)=\frac{5}{8}$ and $P(A\cup B)=\frac{3}{4}$, then $P\left(\frac{B}{\overline{A}}\right)=$

1) $\frac{2}{5}$ 2) $\frac{3}{5}$ 3) $\frac{4}{5}$ 4) $\frac{1}{5}$

70. If A and B are two independent events such that $P(A)=\frac{1}{3}$ and $P(B)=\frac{3}{4}$, then $P\left\{\frac{B}{(A\cup B)}\right\}=$

1) $\frac{7}{10}$ 2) $\frac{8}{10}$ 3) $\frac{9}{10}$ 4) $\frac{6}{10}$

71. E_1, E_2 are events of a sample space such that $P(E_1)=\frac{1}{4}, P\left(\frac{E_2}{E_1}\right)=\frac{1}{2}, P\left(\frac{E_1}{E_2}\right)=\frac{1}{4}$. Then $P\left(\frac{\overline{E_1}}{E_2}\right)=$

1) $\frac{1}{3}$ 2) $\frac{1}{4}$ 3) $\frac{2}{3}$ 4) $\frac{3}{4}$

72. A six - faced unbiased die is thrown twice and the sum of the numbers appearing on the upper face is observed to be 7. The probability that the number 3 has appeared at least once is [EAMCET 2014]

1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{1}{4}$ 4) $\frac{1}{5}$

73. The results of students of a college revealed the following facts. 25% of students failed in Mathematics, 15% of students failed in Chemistry, 10% of students failed in both. If a student is selected at random. The probability that he has failed in Mathematics, given that he failed in Chemistry is

1) $\frac{1}{5}$ 2) $\frac{2}{3}$ 3) $\frac{3}{5}$ 4) $\frac{1}{3}$

74. Suppose E and F are two events of a random experiment. If the probability of occurrence of E is 1/5 and the probability of occurence of F given E is 1/10, then the probability of non-occurrence of at least one of the events E and F is

1) 1/18 2) 1/2 3) 49/50 4) 1/50

75. The probability of drawing 2 red balls in succession from a bag containing 4 red balls and 5 black balls when the ball that is drawn first is not replaced is

1) $\frac{1}{6}$ 2) $\frac{16}{81}$ 3) $\frac{3}{8}$ 4) $\frac{4}{9}$

76. A box contains 5 black, 4 white and 6 red balls. Two balls are drawn without replacement. The probability that the first will be white and the second will be black is

1) $\frac{1}{21}$ 2) $\frac{2}{21}$ 3) $\frac{3}{21}$ 4) $\frac{4}{21}$

PROBLEMS ON INDEPENDENT EVENTS

77. If A and B are two events such that $P(A\cup B)=\frac{5}{6},\ P(A\cap B)=\frac{1}{3}, P(A)=\frac{2}{3},$ then A and B are
1) dependent events
2) independent events
3) mutually exclusive events
4) mutually exclusive and independent events

78. If A and B are two independent events such that $P(B)=\frac{2}{7}$, $P(A\cup B^c)=0.8$ then P(A)=
1) 0.1 2) 0.2 3) 0.3 4) 0.4

79. If A and B are two independent events, and $P(A)=1/4$; $P(B)=1/3$ then, $P((A-B)\cup(B-A))=$
1) $\frac{5}{12}$ 2) $\frac{7}{12}$ 3) $\frac{11}{12}$ 4) $\frac{1}{3}$

80. A and B are two candidates seeking admission in I.I.T. The probability that both A and B are selected is at most 0.4. If the probability of A's selection is 0.5, then the probability of B's selection if A and B are independent is
1) 0.6 2) <0.6 3) ≤ 0.6 4) >0.6

81. A speaks truth in 75% of the cases and B in 80% of the cases. The percentage of cases they are likely to contradict each other in making the same statement is
1) 25% 2) 35% 3) 50% 4) 65%

82. 7 coupons are numbered 1 to 7. Four are drawn one by one with replacement. The probability that the least number appearing on any selected coupon is greater than or equal to 5 is
1) $\left(\frac{3}{7}\right)^4$ 2) $\frac{6}{7^3}$ 3) $\frac{4}{7^3}$ 4) $\left(\frac{3}{4}\right)^4$

83. A box contains 10 mangoes out of which 4 are rotten. Two mangoes are taken together. If one of them is found to be good, the probability that the other is also good is
1) $\frac{1}{3}$ 2) $\frac{8}{13}$ 3) $\frac{5}{13}$ 4) $\frac{2}{3}$

84. Counters numbered 1, 2, 3 are placed in a bag and one is drawn at random and replaced. The operation is being repeated three times. The probability of obtaining a total of 6 is
1) $\frac{7}{9}$ 2) $\frac{6}{27}$ 3) $\frac{7}{27}$ 4) $\frac{5}{27}$

85. A fair die is tossed twice. The probability of getting a 4, 5 or 6 on the 1st toss and a 1, 2, 3 or 4 on the 2nd toss is
1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{1}{4}$ 4) $\frac{1}{8}$

86. An ordinary die has 4 blank faces, one face marked 2 and an other marker 3 Then the probability of obtaining a total of exactly 5 in 2 throws is
1) $\frac{1}{36}$ 2) $\frac{1}{18}$ 3) $\frac{1}{9}$ 4) $\frac{1}{72}$

87. A card is drawn from a well shuffled pack of 52 playing cards. It is replaced in the pack after noting its colour. Again a card is drawn at random. The probability that the 1st card drawn may be a heart and the second card drawn may not be a queen is
1) $\frac{5}{13}$ 2) $\frac{4}{13}$ 3) $\frac{3}{13}$ 4) $\frac{2}{13}$

88. A pack of cards is distributed to four players as in the game of bridge. The probability that a particular player will not get an ace in three consecutive games is
1) $\frac{3\times{}^{48}C_{13}}{{}^{52}C_{13}}$ 2) $\left(\frac{{}^{48}C_{13}}{{}^{52}C_{13}}\right)^3$ 3) $\frac{{}^{48}C_{13}}{{}^{52}C_{13}}$ 4) $\left(\frac{3\times{}^{48}C_{13}}{{}^{52}C_{13}}\right)^3$

89. The probability of drawing 4 white and 2 black balls in two drawings in succession from a bag containing 1 red, 4 black and 6 white balls, if the drawing is without replacement is
1) $\frac{{}^6C_4}{{}^{11}C_4}\times\frac{{}^4C_2}{{}^{11}C_2}$ 2) $\frac{{}^6C_4}{{}^{11}C_4}\times\frac{{}^4C_2}{{}^7C_2}$ 3) $\frac{{}^{10}C_6}{{}^{11}C_6}$ 4) $\frac{{}^{10}C_2}{{}^{11}C_2}$

90. The odds against A solving a certain problem are 4 to 3 and the odds in favour of B solving the same problem are 7 to 5. If both of them try independently, the probability that the problem will be solved is
1) $\frac{11}{21}$ 2) $\frac{13}{21}$ 3) $\frac{16}{21}$ 4) $\frac{8}{21}$

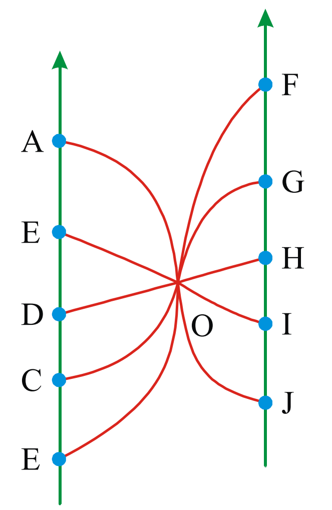
ADDITIONAL PROBLEMS

91. A committee of 6 persons is to be formed from a group of 7 gents and 4 ladies. What is the chance that the committee consists of ladies majority
1) $\frac{7}{22}$ 2) $\frac{3}{22}$ 3) $\frac{5}{22}$ 4) $\frac{1}{22}$

92. If the letters of the word NALGONDA are arranged in arbitrary order, the probability that the letter G, O, D appear in that order is

1) $\frac{1}{6}$ 2) $\frac{1}{24}$ 3) $\frac{1}{120}$ 4) $\frac{1}{10}$

93. The adjoining figure gives the road plan of lanes connecting two parallel roads AB and FJ. A man walking on the road AB takes a turn at random to reach the road FJ. It is known that he reaches the road FJ from O by taking a straight line. The chance that he moves on a straight line from the road AB to the road FJ is.



1) $\frac{1}{5}$ 2) $\frac{2}{5}$ 3) $\frac{3}{5}$ 4) $\frac{4}{5}$

94. A is a set containing n elements. A subset P_1 of A is chosen at random. The set A is reconstructed by replacing the elements of P_1. A subset P_2 is again chosen at random. The probability that $P_1 \cup P_2$ contains exactly one element, is

1) $\frac{3n}{4^n}$ 2) $\frac{3^n}{4^n}$ 3) $\frac{3}{4}$ 4) $\frac{3}{4^n}$

95. Three of the six vertices of a regular hexagon are chosen at random. The probability that the triangle with three vertices is equilateral is

1) $\frac{1}{2}$ 2) $\frac{1}{5}$ 3) $\frac{1}{10}$ 4) $\frac{1}{20}$

LEVEL-I (C.W.) - KEY

1) 2	2) 1	3) 3	4) 3	5)4	6) 2	7) 3
8) 3	9) 4	10) 2	11) 4	12) 3	13) 1	14) 4
15) 1	16) 3	17) 3	18) 2	19) 3	20) 1	21) 2
22) 3	23) 2	24) 3	25) 3	26) 3	27) 3	28) 3
29) 2	30) 2	31) 3	32) 1	33) 3	34) 1	35) 3
36) 3	37) 3	38) 3	39) 1	40) 3	41) 3	42) 2
43) 3	44) 2	45) 1	46) 1	47) 2	48) 2	49) 2
50) 3	51) 2	52) 3	53) 3	54) 1	55) 3	56) 4
57) 1	58) 1	59) 1	60) 2	61) 3	62) 4	63) 3
64) 2	65) 2	66) 3	67) 3	68) 1	69) 2	70) 3
71) 4	72) 2	73) 2	74) 3	75) 1	76) 2	77) 2
78) 3	79) 1	80) 3	81) 2	82) 1	83) 3	84) 3
85) 2	86) 2	87) 3	88) 2	89) 2	90) 3	91) 4
92) 1	93) 1	94) 1	95) 3			

LEVEL-I (C.W.) - HINTS

1. $P(A)+P(B)=1$; $P(A)=\frac{1}{3}$

2. $n(S)={}^8C_1.{}^7C_1$; $E=\{(6, 8), (8,6), (7,8), (8,7)\}$
n(E)=4

3. $n(S)={}^{12}C_4, n(E)={}^6C_2, P(E)=\frac{1}{33}$

4. Req. Prob = $1-\frac{{}^2c_2}{{}^4c_2}=1-\frac{1}{6}=\frac{5}{6}$

5. P(room is lighted) = 1 – P(room is not lightned)
$=1-\frac{{}^4c_3}{{}^{10}c_3}=1-\frac{1}{30}=\frac{29}{30}$

6. $n(S)=7^5$ $n(E)={}^7P_5$

7.P(not consecutive) = $1-P(\text{consecutive})=1-\frac{(40-4+1)}{40_{C_4}}$

8. n(S)=${}^8C_2=28$; A=
$\{(1,2),(1,3),....,(1,8),(2,3),(2,4),....,(2,8),(3,4),(3,5),....,(3,8)\}$
$\therefore n(A)=7+6+5=18.$
$\therefore P(A)=\frac{n(A)}{n(S)}=\frac{18}{28}=\frac{1}{33}$

9. $P(A)=1-\frac{{}^{31}C_2}{{}^{100}C_2}$

10. Req. Prob. $=\frac{2}{6}=\frac{1}{3}$

11. $P(A)=P(B)=3P(C)$; $P(A)+P(B)+P(C)=1$

$P(C)=\frac{1}{7}$

12. n (S) = $4^4=256$; n (A) = $4^4-4!=232$

$P(A)=\frac{29}{32}$

13. Req. Prob = $\frac{3}{2(2n-1)}$ here $2n=20$

14. $x^2+4x+c=0$ has real roots

$\Rightarrow\ (4)^2-4c\geq 0\Rightarrow 16-4c\geq 0\Rightarrow c\in\{1,2,3,4\}$

15. Required probability $=\frac{{}^{15}c_2\times{}^{5}c_1}{{}^{20}c_3}$

16. $P(E)=1-\frac{1}{2^3}=\frac{7}{8}$

17. $P(E)=\frac{1}{2}$

18. $P(E)=\frac{2^6-1-6}{2^6}=\frac{64-7}{64}=\frac{57}{64}$

19. $P(E)=\frac{4+1}{16}=\frac{5}{16}$

20. P(getting a total of odd no. of points)

= P(two heads, one tail) + P(three tails) = $\frac{1}{2}$

21. $P(A)=\frac{5}{36}$

22. $P(E)=\frac{2}{36}=\frac{1}{18}$

23. $P(x\geq 7)=\frac{7}{12}$

24. $P(E)=P(2+4+6+8+10+12)=\frac{1+3+5+5+3+1}{6^2}$

25. $P(|x-y|=4)=\frac{4}{36}=\frac{1}{9}$

26. $P(E)=\frac{5}{9}$

27. $p(x=17)+p(x=18)=\frac{(3)+(1)}{216}=\frac{4}{216}=\frac{1}{54}$

28. $n(E)=n(2,2,2)+n(3,3,0)+n(3,2,1)=1+\frac{3!}{2!}+3!=10$

$n(s)=6^3$

29. $P(E)=\frac{{}^{4}c_2}{6^4}=\frac{1}{6^3}$

30. $P(E)=\frac{2}{36}=\frac{1}{18}$

31. $P(x=5)+P(x=10)=\frac{4+3}{36}=\frac{7}{36}$

32. $P_1=P(E_1)=\frac{1}{36}$; $P_{30}=P(E_{30})=\frac{2}{36}$

$P_4=P(E_4)=\frac{3}{36}$ $P_6=P(E_6)=\frac{4}{36}$

$\therefore P_1<P_{30}<P_4<P_6$

33. $P(A\cap B)=\frac{1}{2}\times\frac{1}{2}=\frac{1}{4}$

34. one even no & 5 odd no's $P(E)=\frac{10}{36}=\frac{5}{18}$

35. Leap - year contains 366 days.

36. The February of leap year contains 29 days

37. Probability to select a calender month $=\frac{1}{12}$

$P(E)=\frac{1}{12}\times\frac{1}{7}=\frac{1}{84}$

38. $P(A)=\frac{3}{7}$

39. For Leap year 366 = 52(7) + 2

1 day	II day
sun	Mon

Req. Prob. is $\frac{1}{2}$

40. Req. prob = $\frac{{}^{13}c_5\times{}^{13}c_3\times{}^{13}c_3\times{}^{13}c_2}{{}^{52}c_{13}}$

41. $P(\text{spade or Ace})=\frac{4}{13}$

42. Req. prob. = $\frac{{}^{13}c_1\times{}^{13}c_1\times{}^{13}c_1\times{}^{13}c_1}{{}^{52}c_4}$

43. $P(E)=\frac{10^4}{{}^{40}c_4}$

44. $P(E)=\frac{{}^{26}C_{13}}{{}^{52}C_{13}}$

45. $P(E)=\dfrac{{}^{3}c_1\times{}^{13}c_1\times 2}{52\times 51}$

46. $n(A)={}^{2}C_1.{}^{50}C_1,\ n(S)={}^{52}C_2$

47. If A is the given event then n(A) = 7! ${}^{8}p_3$ and n(S) = 10 !

48. $n(S)={}^{40}c_2=\dfrac{40\times 39}{2}$; $n(E)={}^{16}c_1\times{}^{14}c_1$

$P(E)=\dfrac{16.14}{\frac{40.39}{2}}$

49. $n(S)={}^{9}C_3;\ n(E)={}^{3}C_1.{}^{4}C_1.{}^{2}C_1$

50. $n(S)={}^{12}c_3$

$n(E)={}^{4}c_2\times{}^{8}c_1+{}^{3}c_2\times{}^{9}c_1+{}^{5}c_2\ {}^{7}c_1$

51. $P(E)=\dfrac{{}^{5}c_3}{{}^{9}c_3}$; $P(\overline{E})=1-\dfrac{{}^{5}c_3}{{}^{9}c_3}$

$R.P=P(\overline{E}):P(E)$

52. $R.P=\dfrac{1}{2}\left[\dfrac{2}{5}+\dfrac{4}{6}\right]$

53. Req. Prob. $=\dfrac{3}{5}\times\dfrac{3}{6}+\dfrac{2}{5}\times\dfrac{4}{6}=\dfrac{17}{30}$

54. $P(A)=\dfrac{{}^{32}c_3}{{}^{64}c_3}$

55. $P(A)=\dfrac{{}^{2}c_1\times{}^{32}c_2\times{}^{32}c_1}{{}^{64}c_3}$

56. $A\subset B\Rightarrow A\cup B=B\Rightarrow P(A\cup B)=P(B)$

57. $P(B-A)=P(B)-P(A\cap B)$

$\Rightarrow P(B)=P(B-A)+P(A\cap B)$

58. P(neither A nor B occurs) $=P(\overline{A}\cap\overline{B})$

$=1-P(A\cup B)$ (By Demorgan law)

$=1-\{P(A)+P(B)-P(A\cap B)\}$

59. $P(A)+P(B)+P(C)=1$

$\Rightarrow P(A)+\dfrac{3}{2}P(A)+\dfrac{1}{2}P(B)=1$

$\Rightarrow P(A)+\dfrac{3}{2}P(A)+\dfrac{1}{2}\left(\dfrac{3}{2}P(A)\right)=1$

60. $P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$\Rightarrow P(B)=P(A\cup B)-P(A)+P(A\cap B)$

$=P(A\cup B)-0.5+0.3\le 1-0.2=0.8$

$P(B)\le 0.8$

61 $P(A)=\dfrac{3}{10},\ P(B)=\dfrac{4}{10}$

$P(A)+P(B)+P(C)=1$

$P(C)=\dfrac{3}{10}$

$P(\overline{C}):P(C)=7:3$

62. $P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$=\dfrac{25}{100}+\dfrac{16}{100}-\dfrac{8}{100}=\dfrac{33}{100}$

63. $P(A)=\dfrac{8}{20},\ P(B)=\dfrac{7}{20}$; $P(\overline{A}\cap\overline{B})=\dfrac{8}{20}$

$P(\overline{A}\cup\overline{B})=P(\overline{A})+P(\overline{B})-P(\overline{A}\cap\overline{B})=\dfrac{17}{20}$

64. $P(A\cup B)=\dfrac{10}{30}+\dfrac{15}{30}-\dfrac{5}{30}=\dfrac{20}{30}=\dfrac{2}{3}$

65. $P(E)=1-pq$

66. $P(A)=P(x\ge 20)=\dfrac{3}{5}$; $P(B)=P(x\ge 20)=\dfrac{2}{3}$

$P(A\cap\overline{B})+P(\overline{A}\cap B)=\dfrac{3}{5}.\dfrac{1}{3}+\dfrac{2}{5}.\dfrac{2}{3}=\dfrac{7}{15}$

67. $P(H)=\dfrac{1}{7},\ P(W)=\dfrac{1}{5}$; $P(H\cap W)=\dfrac{1}{35}$

68. $P(A)=\dfrac{3}{6},\ P(B)=\dfrac{2}{4},\ P(C)=\dfrac{4}{4}$

$P(A\cup B\cup C)=1-P(\overline{A}).P(\overline{B}).P(\overline{C})=1$

69. $P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$\dfrac{3}{4}=\dfrac{3}{8}+\dfrac{5}{8}-P(A\cap B)$; $P(A\cap B)=\dfrac{1}{4}$

$P\left(\dfrac{B}{\overline{A}}\right)=\dfrac{P(\overline{A}\cap B)}{P(\overline{A})}=\dfrac{P(B)-P(A\cap B)}{P(\overline{A})}$

$=1-\dfrac{\frac{1}{4}}{\frac{5}{8}}=\dfrac{3}{5}$

70. $P\left(\dfrac{B}{A\cup B}\right)=\dfrac{P(B)}{P(A\cup B)}=\dfrac{9}{10}$

71. $P\left(\frac{\overline{E_1}}{E_2}\right)=\frac{P(E_2\cap\overline{E_1})}{P(E_2)}=\frac{P(E_2)-P(E_1\cap E_2)}{P(E_2)}$

$=1-\frac{P(E_1\cap E_2)}{P(E_2)}=1-\frac{1}{4}=\frac{3}{4}$

72. $A=\{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$,

$A\cap B=\{(3,4),(4,3)\}$ $P(B/A)=\frac{n(A\cap B)}{n(A)}$

73. $P\left(\frac{M}{C}\right)=\frac{P(M\cap C)}{P(C)}$

74. $P(E)=\frac{1}{5}, P\left(\frac{F}{E}\right)=\frac{1}{10}$

$P(\overline{E}\cup\overline{F})=1-P(E\cap F)=1-\frac{1}{5}.\frac{1}{10}=\frac{49}{50}$

75. $P(A\cap B)=P(A)P\left(B/A\right)=\frac{4}{9}\times\frac{3}{8}=\frac{1}{6}$

76. Required probability= $P(A\cap B)=P(A).P(B/A)$

$=\left(\frac{4}{15}\right)\left(\frac{5}{14}\right)=\frac{2}{21}$

77. $P(A\cup B)+P(A\cap B)=P(A)+P(B)$

$P(B)=\frac{1}{2}, P(A)=\frac{2}{3}$

$\therefore P(A\cap B)=P(A)P(B)$

78. $P(B)=\frac{2}{7},\ P(\overline{B})=\frac{5}{7}\quad P(A\cup\overline{B})=0.8$

$P(A)+P(\overline{B})-P(A).P(\overline{B})=0.8$

$\Rightarrow P(A)+P(B).P(\overline{A})=0.8$

$P(A)=0.3$

79. $P[(A-B)\cup(B-A)]$

= probability that exactly one of A and B happens

$=P(A-(A\cap B))+P(B-(A\cap B))$

$=\{P(A)-P(A\cap B)\}+\{P(B)-P(A\cap B)\}$

$=P(A)+P(B)-2P(A\cap B)$

$=P(A)+P(B)-2P(A)P(B)$

80. $P(A\cap B)\le 0.3$; $\quad P(A)=0.5$

$(0.5)P(B)\le 0.3\quad P(B)\le 0.6$

81. $P(A\cap\overline{B})\cup(\overline{A}\cap B)=P(A).P(\overline{B})+P(\overline{A}).P(B)$

82. $P(x\ge 5)=\frac{3}{7}.\frac{3}{7}.\frac{3}{7}.\frac{3}{7}=\left(\frac{3}{7}\right)^4$

83. Req. Prob. $=\frac{{}^6c_2}{{}^6c_2+{}^6c_1\times{}^4c_1}$

84. (2 2 2) (1 2 3)

Req. prob $=\frac{1}{3}\frac{1}{3}\frac{1}{3}+6\left(\frac{1}{3}\frac{1}{3}\frac{1}{3}\right)=\frac{7}{27}$

85. Req. Prob $=\frac{3}{6}\times\frac{4}{6}=\frac{1}{3}$

86. Req. Prob $=2\times\frac{1}{6}\times\frac{1}{6}=\frac{1}{18}$

87. Req. Prob $=\frac{13}{52}\times\frac{48}{52}=\frac{3}{13}$

88. Req. Prob $=\left(\frac{{}^{48}c_{13}}{{}^{52}c_{13}}\right)^3$

89. Req. Prob $=\frac{{}^6c_4}{{}^{11}c_4}\times\frac{{}^4c_2}{{}^7c_2}$

90. $P(\overline{A}):P(A)=4:3$; $\quad P(B):P(\overline{B})=7:5$

$P(A\cup B)=1-P(\overline{A}).P(\overline{B})=1-\frac{4}{7}.\frac{5}{12}=\frac{16}{21}$

91. $P(E)=\frac{{}^4C_4.{}^7C_2}{{}^{11}C_6}=\frac{1}{22}$

92. $n(E)={}^8P_5,\ n(S)=8!$

93. $n(E)=2,\ n(S)=10$

94. Any element of A has four possibilities : element belongs to (i) both P_1 and P_2 (ii) neither P_1 nor P_2 (iii) P_1 but not to P_2 (iv) P_2 but not to P_1. Thus $n(S)=4^n$. For the favorable cases, we choose one element in n ways and this element has three choices as (i), (iii) and (iv), while the remaining $n-1$ elements have one choice each, namely (ii).

Hence required probability $=\frac{3n}{4^n}$.

95. $P=\frac{2}{20}=\frac{1}{10}$

LEVEL-I (H.W)

1. Suppose $S = \{1, 2, 3, 4\}$ is the sample space of a random experiment. Suppose $P(1) = x, P(2) = 2x, P(3) = 3x$ and $P(4) = 4x$, where P is a probability function. then x is

1) 0.1 2) 0.2 3) 0.3 4) 0.4

2. The probability that a randomly chosen number from the set of first 100 natural numbers is divisible by 4 is

1) $\frac{5}{24}$ 2) $\frac{3}{4}$ 3) $\frac{1}{2}$ 4) $\frac{1}{4}$

3. Out of 30 consecutive integers, two integers are drawn at random. The probability that their sum is an odd number is

1) $\frac{15}{29}$ 2) $\frac{14}{29}$ 3) $\frac{1}{2}$ 4) $\frac{1}{4}$

4. A single letter is selected at random from the word PROBABILITY. The probability that it is a vowel is

1) $\frac{3}{11}$ 2) $\frac{4}{11}$ 3) $\frac{2}{11}$ 4) $\frac{5}{11}$

5. A page is opened at random from a book containing 600 pages. The probability that the number on the page is a perfect square is

1) $\frac{1}{30}$ 2) $\frac{1}{25}$ 3) $\frac{1}{20}$ 4) $\frac{1}{15}$

6. There are 100 pages in a book. If a page of the book is opened at random, the probability that the number on the page is two digit number made up with the same digit is

1) $\frac{8}{100}$ 2) $\frac{9}{100}$ 3) $\frac{1}{10}$ 4) $\frac{8}{10}$

7. 5 different Engineering, 4 different Mathematics and 2 different Chemistry books are placed in a shelf at random. The probability that the books of each kind are all together is

1) $\frac{\angle 5 \angle 4 \angle 2}{\angle 11}$ 2) $\frac{\angle 3 \angle 5 \angle 4 \angle 2}{\angle 11}$

3) $\frac{\angle 5 \angle 6}{\angle 11}$ 4) $\frac{\angle 4 \angle 2}{\angle 11}$

8. The probability of getting a head, when an unbiased coin is tossed is

1) 0 2) $\frac{1}{2}$ 3) 1 4) 2

9. The probability of getting head or tail, when an unbiased coin is tossed is

1) 0 2) $\frac{1}{2}$ 3) 1 4) 2

10. If two coins are tossed 5 times, the chance that there will be 5 heads and 5 tails is

1) $\frac{45}{256}$ 2) $\frac{120}{256}$ 3) $\frac{63}{256}$ 4) $\frac{30}{256}$

11. A coin is weighted so that head is twice as likely to appear as tail. When such a coin is tossed once the probability of getting tail is

1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{2}{3}$ 4) $\frac{1}{4}$

12. When a fair coin is tossed thrice, the probability of obtaining head at most twice is

1) $\frac{1}{8}$ 2) $\frac{5}{8}$ 3) $\frac{7}{8}$ 4) $\frac{3}{8}$

13. A game consists of tossing a coin 3 times and noting its outcome. A boy wins if all tosses give the same outcome and losses otherwise. The probability that the boy losses the game is

1) $\frac{1}{4}$ 2) $\frac{2}{4}$ 3) $\frac{3}{4}$ 4) $\frac{1}{3}$

14. If 10 coins are tossed, the odds against the event of getting atleast 2 heads is

1) 1013:11 2) 1013:10
3) 10:1013 4) 11:1013

15. If 4 fair coins are tossed once then the probability of getting 2 heads and 2 tails is

1) $\frac{3}{8}$ 2) $\frac{5}{8}$ 3) $\frac{7}{8}$ 4) $\frac{1}{2}$

16. When a perfect die is rolled, the probability of getting a face with 4 points upward is

1) $\frac{4}{6}$ 2) $\frac{3}{6}$ 3) $\frac{2}{6}$ 4) $\frac{1}{6}$

17. When a perfect die is rolled, the probability of getting a face with 4 or 5 points upward is

1) $\frac{1}{3}$ 2) $\frac{2}{3}$ 3) $\frac{1}{2}$ 4) $\frac{1}{4}$

18. When a perfect die is rolled, the probability of getting a face with even number of points upward is

1) $\frac{1}{4}$ 2) $\frac{1}{3}$ 3) $\frac{1}{2}$ 4) $\frac{1}{8}$

19. **When a perfect die is rolled the probability of getting any one of the 6 faces upward is**

1) $\frac{1}{6}$ 2) $\frac{1}{2}$ 3) 1 4) 2

20. **In a throw with a pair of symmetrical dice the probability of obtaining a doublet is**

1) $\frac{1}{6}$ 2) $\frac{2}{3}$ 3) $\frac{1}{4}$ 4) $\frac{1}{2}$

21. **When two symmetrical dice are rolled simultaneously, the probability that both the dice show even numbers is**

1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{1}{4}$ 4) $\frac{1}{8}$

22. **The probability of getting at least an ace when two dice are rolled is**

1) $\frac{11}{36}$ 2) $\frac{25}{36}$ 3) $\frac{1}{6}$ 4) $\frac{1}{8}$

23. **Three symmetrical dice are thrown. The probability that the same number will appear on each of them is**

1) $\frac{1}{216}$ 2) $\frac{1}{36}$ 3) $\frac{35}{36}$ 4) $\frac{1}{35}$

24. **Three symmetrical dice are thrown. The probability of obtaining a sum of 16 points is**

1) $\frac{1}{6}$ 2) $\frac{1}{36}$ 3) $\frac{1}{216}$ 4) $\frac{1}{72}$

25. **Two symmetrical dice are thrown at a time. The probability that they show different faces is**

1) $\frac{5}{6}$ 2) $\frac{1}{36}$ 3) $\frac{25}{36}$ 4) $\frac{1}{18}$

26. **Two dice are rolled simultaneously. The probability that the sum of the two numbers on the dice is a prime number is**

1) $\frac{1}{4}$ 2) $\frac{5}{36}$ 3) $\frac{5}{12}$ 4) $\frac{5}{6}$

27. **When two dice are thrown, the probability of getting equal numbers is**

1) $\frac{1}{6}$ 2) $\frac{1}{2}$ 3) $\frac{1}{4}$ 4) $\frac{1}{3}$

28. **The probability that a leap year will have 53 sundays is**

1) $\frac{1}{7}$ 2) $\frac{2}{7}$ 3) $\frac{3}{7}$ 4) $\frac{4}{7}$

29. **The probability that a leap year will have 53 Sundays or 53 Mondays is**

1) $\frac{1}{7}$ 2) $\frac{2}{7}$ 3) $\frac{3}{7}$ 4) $\frac{4}{7}$

30. **The probability that a non-leap year will have only 52 Fridays is**

1) $\frac{3}{7}$ 2) $\frac{4}{7}$ 3) $\frac{5}{7}$ 4) $\frac{6}{7}$

31. **The chance that a leap year selected at random will contain 53 sundays is**

1) 1/7 2) 1/14 4) 3/7 4) 4/7

32. **Two cards are drawn at random from a pack of 52 well shuffled playing cards. The probability that the cards drawn are aces is**

1) $\frac{5}{221}$ 2) $\frac{3}{221}$ 3) $\frac{1}{221}$ 4) $\frac{4}{221}$

33. **When a card is drawn at random from a well shuffled pack of 52 playing cards, the probability that it may be either king or queen is**

1) $\frac{8}{13}$ 2) $\frac{5}{13}$ 3) $\frac{2}{13}$ 4) $\frac{1}{13}$

34. **If two cards are drawn from a well shuffled pack of 52 playing cards, the probability that there will be at least one club card is**

1) $\frac{{}^{39}c_2}{{}^{52}c_2}$ 2) $1-\frac{{}^{39}c_2}{{}^{52}c_2}$ 3) $\frac{39}{52}\times\frac{39}{52}=\frac{9}{16}$ 4) $\frac{{}^{30}c_2}{{}^{52}c_2}$

35. **From a well shuffled pack of 52 playing cards two cards are drawn at random. The probability that either both are red or both are kings is**

1) $\frac{\left({}^{26}c_2+{}^{4}c_2\right)}{{}^{52}c_2}$ 2) $\frac{\left({}^{26}c_2+{}^{4}c_2-{}^{2}c_2\right)}{{}^{52}c_2}$ 3) $\frac{{}^{30}c_2}{{}^{52}c_2}$ 4) $\frac{{}^{39}c_2}{{}^{52}c_2}$

36. **From a well shuffled pack of 52 playing cards, four are drawn at random. The probability that all are spades, but one is a king is**

1) $\frac{{}^{39}c_4}{{}^{52}c_4}$ 2) $\frac{{}^{12}c_3}{{}^{52}c_4}$ 3) $1-\frac{{}^{39}c_4}{{}^{52}c_4}$ 4) $\frac{{}^{12}c_4}{{}^{52}c_4}$

37. **Two cards are drawn from a well shuffled pack of 52 playing cards. The probability that they belong to different colours is**

1) $\frac{2\times{}^{13}c_2}{{}^{52}c_2}$ 2) $\frac{{}^{4}c_2\times13\times13}{{}^{52}c_2}$

3) $\frac{{}^{26}c_1\times{}^{26}c_1}{{}^{52}c_2}$ 4) $\frac{{}^{13}c_2}{{}^{52}c_2}$

38. A card is drawn from a well shuffled pack of 52 cards numbered 2 to 54 The probability that the number on the card is a prime less than 10 is

1) $\frac{4}{13}$ 2) $\frac{3}{13}$ 3) $\frac{2}{13}$ 4) $\frac{1}{13}$

39. Two cards are drawn from a well shuffled pack of 52 playing cards. The probability that one is a heart card and the other is not a heart card is

1) $\frac{7}{34}$ 2) $\frac{9}{34}$ 3) $\frac{13}{34}$ 4) $\frac{15}{34}$

40. A card is drawn at random from a well shuffled pack of 52 cards. Again a card is drawn at random from the remaining cards. The probability that one is a king and the other is a queen is

1) $\frac{4}{663}$ 2) $\frac{8}{663}$ 3) $\frac{1}{221}$ 4) $\frac{2}{663}$

41. The probability of drawing a card which is at least a spade or a king from a well shuffled pack of cards is

1) $\frac{1}{4}$ 2) $\frac{1}{13}$ 3) $\frac{4}{13}$ 4) $\frac{2}{13}$

42. An urn contains 25 balls numbered 1 to 25. Two balls drawn one at a time with replacement. The probability that both the numbers on the balls are odd is

1) $\frac{^{13}c_2}{625}$ 2) $\frac{169}{625}$ 3) $\frac{^{25}c_2}{625}$ 4) $\frac{139}{625}$

43. A bag contains 10 balls out of which two are red, three are blue and five are black. Three balls are drawn at random from the bag. The probability that the balls are of the same colour is

1) $\frac{9}{120}$ 2) $\frac{11}{120}$ 3) $\frac{13}{120}$ 4) $\frac{17}{120}$

44. 3 red and 4 white balls of different sizes are arranged in a row at random. The probability that no two balls of the same colour are together is

1) $\frac{6}{35}$ 2) $\frac{3}{35}$ 3) $\frac{1}{35}$ 4) $\frac{9}{35}$

45. A bag contains 6 white and 4 black balls. Two balls are drawn at random. The probability that they are of the same color is

1)1/15 2)2/5 3)4/15 4)7/15

46. Seven balls are drawn simultaneously from a bag containing 5 white and 6 green balls. The probability of drawing 3 white and 4 green balls is

1) $\frac{7}{^{11}c_7}$ 2) $\frac{^5c_3+{}^6c_4}{^{11}c_7}$ 3) $\frac{^5c_2\times{}^6c_2}{^{11}c_7}$ 4) $\frac{^6c_3\times{}^5c_4}{^{11}c_7}$

47. If two balls are drawn from a bag containing 3 white, 4 black and 5 red balls, then the probability that the drawn balls are of different colours is

1) $\frac{60}{66}$ 2) $\frac{47}{66}$ 3) $\frac{12}{60}$ 4) $\frac{13}{60}$

48. A bag contains 3 red, 4 white and 7 black balls. The probability of drawing a red or a black ball is

1) $\frac{2}{7}$ 2) $\frac{5}{7}$ 3) $\frac{3}{7}$ 4) $\frac{4}{7}$

49. Three '1×1' squares of a chess board having 8×8 squares are chosen at random, the chance that all the three of the same colour is

1) $\frac{^{32}c_3}{^{64}c_3}$ 2) $\frac{2\times{}^{32}c_3}{^{64}c_3}$ 3) 1 4) $\frac{^4c_3}{^{64}c_3}$

50. If A and B are events of a random experiment such that $P(A\cup B)=4/5, P(\overline{A}\cup\overline{B})=7/10$ $P(B)=2/5,$ then P(A) = [EAMCET 2009)

1) $\frac{9}{10}$ 2) $\frac{8}{10}$ 3) $\frac{7}{10}$ 4) $\frac{3}{5}$

51. If A and B are two events such that $P(A\cup B)=\frac{3}{4}, P(A\cap B)=\frac{1}{4}$ & $P(\overline{A})=\frac{2}{3},$ then $P(\overline{A}\cap B)=$

1) $\frac{1}{12}$ 2) $\frac{2}{12}$ 3) $\frac{7}{12}$ 4) $\frac{5}{12}$

52. In a swimming competition, only 3 students *A, B* and *C* are taking part. The probability of *A*'s winning or the probability of *B*'s winning is three times the probability of *C*'s winning. The probability of the event either *B* or *C* to win is

1) $\frac{1}{7}$ 2) $\frac{2}{7}$ 3) $\frac{3}{7}$ 4) $\frac{4}{7}$

53. Two events A and B have probability 0.25 and 0.50 respectively. The probability that both A andB occur simultaneously is 0.14. Then the probability that neither A nor B occurs is

1) 0.39 2) 0.25 3) 0.89 4) 0.50

54. In a class of 60 boys and 20 girls, half of the boys and half of the girls know cricket, then the probability of the event that a person selected from the class is either a boy, or a girl who knows cricket is

1) $\frac{1}{2}$ 2) $\frac{3}{8}$ 3) $\frac{5}{8}$ 4) $\frac{7}{8}$

55. In a class of 125 students, 70 passed in mathematics, 55 passed in statistics and 30 in both. The probability that a student selected at random from that class has passed in only one subject

1) $\frac{13}{25}$ 2) $\frac{3}{25}$ 3) $\frac{17}{25}$ 4) $\frac{8}{25}$

56. The probability that a company executive will travel by train is 2/3 and that he will travel by plane is 1/5. The probability of his travelling by train or plane is

1) $\frac{2}{15}$ 2) $\frac{13}{15}$ 3) $\frac{15}{13}$ 4) $\frac{15}{2}$

57. There are 5 green, 6 black and 7 white balls in abag. A ball is drawn at random from the bag. The probability that it may be either green or black is

1) $\frac{5}{18}$ 2) $\frac{6}{18}$ 3) $\frac{11}{18}$ 4) $\frac{13}{18}$

58. A bag contains 25 balls numbered 1 to 25. One ball is drawn at random. The probability that the number on the ball drawn will be a multiple of 5 or 6 is

1) $\frac{3}{25}$ 2) $\frac{7}{25}$ 3) $\frac{9}{25}$ 4) $\frac{5}{25}$

59. Suppose there are 12 boys and 4 girls in a class. If we choose three children one after another in succession at random, the probability that all the three are boys is

1) $\frac{5}{28}$ 2) $\frac{11}{28}$ 3) $\frac{9}{28}$ 4) $\frac{3}{4}$

60. A perfect die is rolled. If the outcome is an odd number, the probability that it is a prime is

1) $\frac{1}{3}$ 2) $\frac{2}{3}$ 3) $\frac{1}{2}$ 4) $\frac{1}{8}$

61. Two dice are thrown at a time and the sum of the numbers on them is 6. The probability of getting the number 4 on any one of them is

1) $\frac{2}{5}$ 2) $\frac{1}{5}$ 4) $\frac{2}{3}$ 4) $\frac{1}{3}$

62. A die is thrown 3 times. The probability of the event of getting sum of the numbers thrown as 15 when it is known that the first throw was a five is

1) $\frac{1}{36}$ 2) $\frac{2}{36}$ 3) $\frac{3}{36}$ 4) $\frac{4}{36}$

63. An urn contains 12 red balls and 12 green balls. Suppose two balls are drawn one after another without replacement, then the probability that the second ball drawn is green given that the first ball drawn is red is

1) $\frac{6}{23}$ 2) $\frac{12}{23}$ 3) $\frac{11}{23}$ 4) $\frac{17}{23}$

64. If A and B are two events such that $P(A)=\frac{1}{4}$ **and** $P(A\cup B)=\frac{1}{3}$ **and P(B)=P, the value of P if A and B are independent**

1) $\frac{1}{9}$ 2) $\frac{2}{9}$ 3) $\frac{4}{9}$ 4) $\frac{5}{9}$

65. If A and B are independent events of a random experiment such that $P(A\cap B)=\frac{1}{6}$ **and** $P(\overline{A}\cap\overline{B})=\frac{1}{3}$ **then** $P(A)=$

1) 1/4, 1/3 2) 1/3, 1/2
3) 1/2, 1/5 4) 2/3, 1/3

66. A and B are two events such that $P(A)=0.4,$ $P(A\cup B)=0.7$ **. If A and B are independent then P(B) =**

1) 0.3 2) 0.4 3) 0.5 4) 0.7

67. A problem is given to 3 students. Their chances of solving it individually are $\frac{1}{3},\frac{1}{4}$ **and** $\frac{1}{5}$ **. The probability that the problem will be solved is**

1) $\frac{2}{5}$ 2) $\frac{3}{5}$ 3) $\frac{4}{5}$ 4) $\frac{1}{5}$

68. In a shooting test the probability of *A,B,C* hitting the targets are 1/2 , 2/3 and 3/4 respectively. If all of them fire at the same target. The probability that only one of them hits the target is

1) $\frac{1}{5}$ 2) $\frac{1}{2}$ 3) $\frac{1}{3}$ 4) $\frac{1}{4}$

69. A person is known to hit the target in 3 out of 4 shots, where as an other person is known to hit twice in every three attempts. If both of them try independently the probability that the target being hit is

1) $\frac{1}{12}$ 2) $\frac{11}{12}$ 3) $\frac{5}{12}$ 4) $\frac{3}{12}$

70. A speaks truth in 75% of the cases and B in 80% of the cases. The percentage of cases they are likely to concur with each other in making the same statement is

1) 25% 2) 35% 3) 50% 4) 65%

71. The probability that a student passes in Mathematics is $\frac{2}{3}$ and the probability that he passes in English is $\frac{4}{9}$. The probability that he passes in any one of the courses is $\frac{4}{5}$. The probability that he passes in both is

1) $\frac{11}{45}$ 2) $\frac{14}{45}$ 3) $\frac{17}{45}$ 4) $\frac{15}{45}$

72. Three mangoes and three apples are in a box. If two fruits are chosen at random, the probability that one is a mango and the other is an apple is

1) $\frac{3}{5}$ 2) $\frac{5}{6}$ 3) $\frac{1}{36}$ 4) $\frac{1}{6}$

73. A bag contains 17 counters marked with numbers 1 to 17 on chits. A counter is drawn and replaced. A second draw is then made. The chance that the number on the counter drawn 1st is even and the second is odd is

1) $\frac{64}{289}$ 2) $\frac{81}{289}$ 3) $\frac{72}{289}$ 4) $\frac{36}{289}$

74. Two cards are selected at random from 10 cards numbered 1 to 10. The probaility that their sum is odd, if the 2 cards are drawn one after an other without replacement is

1) $\frac{5}{9}$ 2) $\frac{4}{9}$ 3) $\frac{2}{9}$ 4) $\frac{3}{9}$

75. A and B are two events such that P(A)=0.5, $P\left(A\cup\bar{B}\right)=0.7$. If A and B are independent events P(B) =

1)5 2)5 3) $\frac{3}{5}$ 4) $\frac{4}{5}$

76. There are 10 cards in a bag. On 5 of them "N" is printed and on the other 5 "C" is printed. 3 cards are drawn, one after an other without replacement and kept in that order. The probability that the word formed with the letters is NCC is

1) $\frac{5}{36}$ 2) $\frac{7}{36}$ 3) $\frac{11}{36}$ 4) $\frac{3}{36}$

77. For an oral test 25 questions are set of which 5 are easy and 20 are tough. Two questions are given to two candidates A and B in that order (one question to each person). The probability for B to receive easy question is

1) $\frac{1}{5}$ 2) $\frac{4}{5}$ 3) $\frac{5}{24}$ 4) $\frac{19}{24}$

78. Three faces of a fair die are yellow, two faces red and one blue. The die is tossed 3 times. The probability that the colours yellow, red and blue appear in the 1st, 2nd and 3rd tosses respectively is

1) $\frac{1}{6}$ 2) $\frac{1}{36}$ 3) $\frac{35}{36}$ 4) $\frac{1}{3}$

79. The chance of throwing an ace in the 1st only, of the two successive throws with an ordinary die is

1) $\frac{1}{6}$ 2) $\frac{1}{36}$ 3) $\frac{5}{36}$ 4) $\frac{5}{6}$

80. From a well shuffled pack of 52 playing cards two cards are drawn at random, one after an other without replacement. The probability that 1st one is a king and second one is queen is

1) $\frac{5}{663}$ 2) $\frac{4}{663}$ 3) $\frac{1}{221}$ 4) $\frac{3}{221}$

81. A bag contains 10 white and 8 black balls. Two successive drawings of 2 balls are made. The probability that the 1st draw will give 2 white and the 2nd draw will give 2 black if the drawing is without replacement is

1) $\frac{{}^{10}c_2+{}^{8}c_2}{{}^{18}c_2}$ 2) $\frac{{}^{10}c_2}{{}^{18}c_2}\times\frac{{}^{8}c_2}{{}^{16}c_2}$

3) $\frac{{}^{10}c_2}{{}^{18}c_2}\times\frac{{}^{8}c_2}{{}^{18}c_2}$ 4) $\frac{{}^{12}c_2}{{}^{18}c_2}\times\frac{{}^{6}c_2}{{}^{18}c_2}$

82. A and B toss a coin alternately till one of them gets a head and wins the game. The probability of A's winning if A starts the game is

1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{2}{3}$ 4) $\frac{1}{4}$

83. A bag contains 4 white and 2 black balls. Another contains 3 white and 5 black balls. If one ball is drawn from each, the probability that both are black is

1) $\frac{13}{24}$ 2) $\frac{5}{24}$ 3) $\frac{1}{14}$ 4) $\frac{2}{14}$

LEVEL-I (H.W.) - KEY

1)1	2) 4	3) 1	4)2	5) 2	6)2	7) 2
8) 2	9) 3	10) 3	11) 2	12) 3	13) 3	14) 4
15)1	16) 4	17) 1	18) 3	19) 3	20) 1	21) 3
22) 1	23) 2	24) 2	25) 1	26) 3	27) 1	28) 2
29) 3	30) 4	31) 2	32) 3	33) 3	34) 2	35) 2
36) 2	37) 3	38) 4	39) 3	40) 2	41) 3	42) 2
43) 2	44) 3	45) 4	46) 3	47) 2	48) 2	49) 2
50) 3	51) 4	52) 4	53) 1	54) 4	55) 1	56) 2
57) 3	58) 3	59)2	60) 2	61) 1	62)3	63) 2
64) 1	65) 2	66) 3	67) 2	68) 4	69) 2	70) 4
71) 2	72) 1	73) 3	74) 1	75) 3	76) 1	77) 1
78) 2	79) 3	80) 2	81) 2	82) 3	83) 2	

LEVEL-I (H.W.) - HINTS

1. Since P is a probability function, we have

$$\sum_{i=1}^{4} P(i) = 1$$
$\Rightarrow P(1) + P(2) + P(3) + P(4) = 1$
$\Rightarrow x + 2x + 3x + 4x = 1$

2. $P(A) = \frac{25}{100} = \frac{1}{4}$

3. $\therefore P(A) = \frac{n(A)}{n(s)} = \frac{15\times15}{^{30}C_2} = \frac{15\times15\times2}{30\times29} = \frac{15}{29}$

4. $P(A) = \frac{4}{11}$

5. $P(A) = \frac{24}{600} = \frac{1}{25}$

6. $n(S) = {}^{100}c_1 = 100$; $n(A) = 9\{(1,1),(2,2),....(9,9)\}$

$P(A) = \frac{9}{100}$

7. $n(A) = \angle 5 \angle 4 \angle 2 \angle 3$; n(S) = $\angle 11$

8. $P(H) = \frac{1}{2}$

9. $P(H\cup T) = P(H) + P(T) - P(H\cap T)$

10. $P(E) = \frac{^{10}c_5}{2^{10}}$

11. P(H)=2P(T) ; $P(H) + P(T) = 1$

12. $P(\text{At most } 2H) = P(2\text{heads}) + P(1\text{head}) + P(0\text{heads})$

13. $\therefore P(A) = 1 - P(\bar{A}) = 1 - \frac{1}{4} = \frac{3}{4}$

14. $n(A^C) = {}^{10}C_0 + {}^{10}C_1 = 11$

$\therefore n(A) = 2^{10} - 11 = 1013$;

15. $P(A) = \frac{n(A)}{n(S)} = \frac{6}{16} = \frac{3}{8}$

16. $P(A) = \frac{1}{6}$

17. $P(A\cup B) = P(A) + P(B) - P(A\cap B)$

18. Req.prob = P(2)+P(4)+P(6)

19. $P(E) = \frac{1}{6}(6) = 1$

20. $P(E) = \frac{1}{6}$

n(doublet)= {(1, 1),(2,2),(3,3),(4,4),(5,5),(6,6)}
=6 ; n(s) = 36

21. $P(E) = \frac{9}{36} = \frac{1}{4}$

22. $P(E) = 1 - \frac{25}{36} = \frac{11}{36}$

23. $P(E) = \frac{6}{6^3} = \frac{1}{36}$

24 $n(E) = 6\,; n(s) = 6^3$

25. $P(E) = \frac{6\times5}{6\times6} = \frac{5}{6}$

26. $P(2+3+5+7+11) = \frac{1+2+4+6+2}{36} = \frac{15}{36} = \frac{5}{12}$

27. $P(E) = \frac{1}{6}$

28. $P(A) = \frac{n(A)}{n(S)} = \frac{2}{7}$

29. Leap year contains 366 days.
It has 52 weeks and remaining 2 days left

30. Non-leap year contains 365 days.
It has 52 weeks and 1 day left

31. Required probability $= \frac{1}{4}\times\frac{2}{7} = \frac{1}{14}$

32. Req. prob $= \frac{^4c_2}{^{52}c_2} = \frac{1}{221}$

33. $P(K\cup Q)=\frac{4}{52}+\frac{4}{52}=\frac{2}{13}$

34. Req. prob $=1-\frac{{}^{39}c_2}{{}^{52}c_2}$

35. P(Red or kings) $=\frac{{}^{26}c_2+{}^{4}c_2-{}^{2}c_2}{{}^{52}c_2}$

36. $P(E)=\frac{{}^{12}c_3\times{}^{1}C_1}{{}^{52}c_4}$

37. $P(E)=\frac{{}^{26}c_1\times{}^{26}c_1}{{}^{52}c_2}$

38. $P(E)=\frac{4}{52}=\frac{1}{13}$

39. $P(E)=\frac{{}^{13}C_1\times{}^{39}C_1}{{}^{52}C_2}$

40. $P(A\cap B)=\frac{8}{663}$

41. $P(A\cup K)=\frac{13}{52}+\frac{4}{52}-\frac{1}{52}=\frac{16}{52}=\frac{4}{13}$

42. $n(S)=25\times25$

$n(E)=13\times13$

$P(E)=\frac{n(E)}{n(S)}$

43. $n(s)={}^{10}c_3$

$n(E)={}^{3}c_3+{}^{5}c_3$

44. $n(S)=7!$

$n(E)=3!\times4!$

45. $n(S)={}^{10}c_2$

$n(E)={}^{6}c_2+{}^{4}c_2$

46. $n(S)={}^{11}c_7$

$n(E)={}^{5}c_3.{}^{6}c_4$

47. $n(S)={}^{12}c_2$

$n(E)={}^{3}c_1\times{}^{4}c_1+{}^{4}c_1\,{}^{5}c_1+{}^{5}c_1\,{}^{3}c_1$

48. $n(S)={}^{14}c_1$

$n(E)={}^{3}c_1+{}^{7}c_1$

49. $P(A)=\frac{{}^{2}c_1\times{}^{32}c_3}{{}^{64}c_3}$

50. $P(\overline{A}\cup\overline{B})=P(\overline{A\cup B})=1-P(A\cap B)$

$=1-[P(A)+P(B)-P(A\cup B)]$

$\Rightarrow\frac{7}{10}=1-P(A)-\frac{2}{5}+\frac{4}{5}$

51. $P(A\cup B)=\frac{3}{4},\ P(A\cap B)=\frac{1}{4},\ P(A)=\frac{1}{3}$

$P(\overline{A}\cap B)=P(B)-P(A\cap B)$

$P(A\cup B)=P(A)+P(B)-P(A\cap B)$

52. $P(B\cup C)=P(B)+P(C)$ $[\because B\cap C=\phi]$

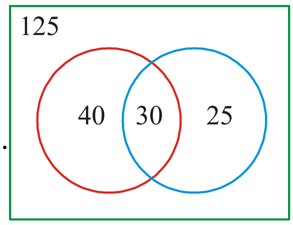
$=\frac{3}{7}+\frac{1}{7}=\frac{4}{7}$

53. $P(\overline{A}\cap\overline{B})=1-P(A\cup B)$

$=1-(0.25+0.50-0.14)=1-0.61$

$=0.39$

54. $P(A\cup B)=\frac{60}{80}+\frac{10}{80}=\frac{7}{8}$

55. 

required probability $=\frac{40}{125}+\frac{25}{125}$

56. $P(A\cup B)=P(A)+P(B)=\frac{2}{3}+\frac{1}{5}=\frac{13}{15}$

57. $n(S)={}^{18}c_1$

$n(E)={}^{5}c_1+{}^{6}c_1$

58. $n(S)={}^{25}c_1$

$n(E)={}^{5}c_1+{}^{4}c_1$

59. Required probability

$=P(A\cap B\cap C)$

$=P(A).P(B/A).P(C/A\cap B)=\frac{12}{16}.\frac{11}{15}.\frac{10}{14}=\frac{11}{28}$

60. Odd no ={ 1,3,5 } ;

$$P\left(\frac{prime\,no}{odd\,no}\right)=\frac{p\left(prime\,nos\,odd\right)}{p\left(odd\,no\right)}=\frac{2}{3}$$

61. $P\left(\frac{A}{B}\right)=\frac{P(A\cap B)}{P(B)}$

62. $P(B/A)=\frac{n(A\cap B)}{n(A)}=3/36=\frac{1}{12}$

63. $P\left(\frac{G}{R}\right)=\frac{P(G\cap R)}{P(R)}$

64. $P(A\cup B)=P(A)+P(B)-P(A)P(B)$

$\therefore P(B)=\frac{1}{9}$

65. $P(A)=x,\ P(B)=y$

$P(A\cap B)=\frac{1}{6}\Rightarrow xy=\frac{1}{6}$

$P(\overline{A}\cap\overline{B})=\frac{1}{3}\Rightarrow(1-x)(1-y)=\frac{1}{6}$

66. $P(A\cup B)=0.7\Rightarrow P(A)+P(B)-P(A).P(B)$

$=0.7$

Find P(B)

67. $P(A\cup B\cup C)=1-P(\overline{A}).P(\overline{B}).P(\overline{C})$

68. Required probability

$$=\frac{1}{2}\times\frac{1}{3}\times\frac{1}{4}+\frac{1}{2}\times\frac{2}{3}\times\frac{1}{4}+\frac{1}{2}\times\frac{1}{3}\times\frac{3}{4}$$

69. $P(A)=\frac{3}{4},\ P(B)=\frac{2}{3}$

$P(A\cup B)=1-P(\overline{A}\cap\overline{B})=1-\frac{1}{4}\times\frac{1}{3}=\frac{11}{12}$

70. $P(A\cap B)\cup(\overline{A}\cap\overline{B})$

$=P(A)P(B)+P(\overline{A}).P(\overline{B})$

71. $P(A\cap B)=\frac{2}{3}+\frac{4}{9}-\frac{4}{5}=\frac{14}{45}$

72. $n(S)={}^{6}c_{2}$

$n(E)={}^{3}c_{1}\times{}^{3}c_{1}$

73. $P(A\cap B)=P(A).P(B)$

74. Req. Prob. $=2\times\frac{5}{10}\times\frac{5}{9}=\frac{5}{9}$

75. $P(A\cup\overline{B})=P(A)+P(\overline{B})-P(A\cap\overline{B})$

$=P(A)+P(\overline{B})P(\overline{A})$

$0.7=0.5+P(\overline{B})0.5$

$P(\overline{B})=\frac{2}{5}\qquad P(B)=\frac{3}{5}$

76. Req. Prob. $=\frac{5}{10}\times\frac{5}{9}\times\frac{4}{8}=\frac{5}{36}$

77. Req. Prob. $=\frac{20}{25}.\frac{5}{25}+\frac{5}{25}.\frac{5}{25}$

78. Req. Prob $=\frac{3}{6}\times\frac{2}{6}\times\frac{1}{6}=\frac{1}{36}$

79. Req. Prob $=\frac{1}{6}\times\frac{5}{6}=\frac{5}{36}$

80. Req. Prob $=\frac{4}{52}\times\frac{4}{51}=\frac{4}{663}$

81. Req. Prob $=\frac{{}^{10}c_{2}}{{}^{18}c_{2}}\times\frac{{}^{8}c_{2}}{{}^{16}c_{2}}$

82. $p=\frac{1}{2},q=\frac{1}{2}$

$p(A)=\frac{p}{1-q^{2}}$

83. $p(E)=\frac{2}{6}\times\frac{5}{8}=\frac{5}{24}$